

The Price of Thought: A Multilingual Analysis of Reasoning, Performance, and Cost of Negotiation in Large Language Models

Anonymous ACL submission

Abstract

This document is a supplements to the general instructions for *ACL authors. It contains instructions for using the L^AT_EX style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers. This document is a supplements to the general instructions for *ACL authors. It contains instructions for using the L^AT_EX style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Introduction

Negotiation is a key aspect of human social and economic behaviour that poses significant challenges for AI systems. The problem is complex, as agents must produce coherent language while performing strategic reasoning, modelling the opponents' preferences and goals, and managing the tension between cooperation and competition to maximise their own outcomes. With LLMs increasingly being tested as autonomous agents in domains ranging from e-commerce to resource distribution, new protocols for agents are being developed, e.g., the Agents to Payments (AP2) protocol (Google Cloud, 2025). It has become essential to assess the limitations and capabilities of language agents, or vanilla LLMs, for negotiation. Negotiation tasks provide a strong test-bed for measuring agent competence,

offering interactive, multi-turn scenarios with objective measures that go beyond static evaluation.

Studies show that LLMs often deviate from optimal play with unreliable performance, where even top-performing systems can lose to weaker opponents and fail in cooperative scenarios (Davidson et al., 2024; Hua et al., 2024; Bianchi et al., 2024; Kwon et al., 2024). These models can acquire deceptive tactics, e.g., expressing false interest in low-value items for later concessions (Lewis et al., 2017), express desperation to improve outcomes (Bianchi et al., 2024), perform much worse as buyers than sellers (Xia et al., 2024), or even take economic risks that lead to budget violations and overpayment (Zhu et al., 2025).

Despite this growing body of research, two critical gaps remain unexplored. First, no previous work has systematically investigated how Chain-of-Thought prompting and reasoning capabilities influence negotiation abilities in terms of both performance and computational cost, a fundamental oversight given that reasoning is central to strategic decision-making. Second, the field has been constrained by monolingual English-only analyses, leaving multilingual negotiation capabilities completely unexplored. To address these significant limitations, we conduct the first comprehensive study examining the effects of reasoning on negotiation tasks across three languages: English, German, and Italian. We implement three distinct dialogue games that require bargaining skills, collaboration, and strategic preference management (von Neumann and Morgenstern, 1944; Nash, 1953; Fisher et al., 2011), testing both commercial and open-weight LLMs in a self-play mode where both sides of the games are played against themselves. We analyse the impact of systematically turning on and off the reasoning mode of evaluated models.

Our investigation targets three fundamental research questions:

RQ1: *What is the computational and perfor-*

mance trade-off between reasoning overhead and negotiation effectiveness across different tasks and languages?

RQ2: To what extent do models maintain language consistency in their reasoning processes when performing multilingual negotiation tasks?

RQ3: Do models demonstrate genuine strategic adaptation over multiple turns, or merely surface-level pattern matching that creates an illusion of thinking?

2 Related Work

Game theory and negotiation research have a deep foundation in the analysis of strategic interactions between rational agents, with seminal contributions from (Nash, 1950) on bargaining solutions, (Rubinstein, 1982) on sequential bargaining, and (Raiffa, 1985) on negotiation analysis. The convergence of Large Language Models (LLMs) and game-theoretic frameworks represents a significant development in the evaluation and understanding of strategic capabilities in AI systems (Sun et al., 2025). Negotiation games serve as demanding test environments for LLMs, requiring integration of linguistic competence, strategic planning, and social reasoning capabilities.

Benchmarking strategic capabilities in LLMs: systematic evaluation of LLM negotiation performance reveals substantial variance in strategic behaviour, with models frequently departing from optimal play and exhibiting asymmetric performance patterns (Davidson et al., 2024; Hua et al., 2024; Akata et al., 2023) or when some powerful models underperform against weaker opponents and encounter difficulties in cooperative bargaining scenarios. This has motivated the creation of comprehensive evaluation frameworks and experimental platforms designed to assess agent (initialised as LLMs) behaviour across various negotiation contexts, including resource allocation and pricing tasks (Bianchi et al., 2024; Xia et al., 2024). Empirical studies reveal that LLMs can display irrational tactics, e.g. deceptive strategies such as expressing false interest in low-value items to create bargaining leverage in subsequent exchanges (Lewis et al., 2017). LLMs also exhibit susceptibility to cognitive biases, including anchoring effects and responsiveness to social manipulation tactics, where expressions of desperation or aggressive language can substantially affect negotiation outcomes (Bianchi et al., 2024). However, these behaviours present

practical risks, including budget constraint violations and acceptance of economically disadvantageous agreements (Zhu et al., 2025). A consistent finding across studies is the asymmetric difficulty in buyer versus seller roles, with agents systematically underperforming as buyers (Xia et al., 2024).

Improving strategic reasoning and decision-making: Given the inconsistent strategic reasoning observed in current LLMs (Wong et al., 2025), research efforts focus on enhancing decision-making through structured approaches such as integration of game-theoretic solvers with LLM dialogue (Gemp et al., 2024), structured reasoning workflows based on Dominant Strategy Search and Backward Induction (Hua et al., 2024), or even learn by interacting based with reinforcement learning (Cao et al., 2018). From prompting perspective, Chain-of-Thought reasoning emerges as a critical factor in agent performance (Gandhi et al., 2023), contributing to more consistent outcomes in integrative negotiation settings (Vaccaro et al., 2025). Hybrid architectural approaches have been developed to constrain agent behaviour, such as employing deterministic offer generation modules to control pricing decisions while utilising LLMs for natural language dialogue production, resulting in substantial improvements in deal completion rates and profit margins (Xia et al., 2024).

3 Evaluating Negotiation Abilities with Dialogue Games

In this section, we provide details on how we implement three dialogue games to test negotiation abilities in LLMs. Each game has its unique properties, specifically measuring bargaining, collaboration, grounding, or maximising personal preference. We define a dialogue game as a structured communication between two agents (Large Language Models, or LLMs) according to a given communication protocol aimed at achieving a defined goal. The goals in games are defined as the expected outcome, e.g., making a deal, that agents attempt to maximise. The communication protocols dictate how messages are expected to be formatted. A negotiation task unfolds by agents starting to communicate with one another for the given goal, and the conversation continues until it reaches one of the defined stopping criteria: 1) the goal state is reached, 2) the maximum turn limit is reached, or 3) agents violate the communication protocol and the game-play is aborted. The orchestration of this

message passing between agents, validation of communication protocols, and checking whether goal states are reached are done by the Game Master, a scripted entity in the dialogue. We use the *clem-bench framework* (Chalamalasetti et al., 2023). The dialogue games are run in a self-play mode where both participants of the game are LLMs playing against or with one another.

Next, we provide details about each specific game.

3.1 Deal or No Deal

The Deal or No Deal (DoND) game is a two-player game (Lewis et al., 2017) designed to simulate a multi-issue bargaining scenario. Figure 1 shows an example of a successfully played episode of the game. Two players must communicate to reach a mutually beneficial agreement on the division of a set of items, each holding different values for each player. The game focuses on evaluating negotiation skills, including the ability to express and understand preferences, as well as compromise. We provide all prompts and other details in Appendix B.

Game Mechanics Two agents negotiate over a shared set of items, each having a different private value for item types. Players exchange free-form messages and then simultaneously make secret proposals using specific syntax. If proposals are compatible (enough items are requested for both), each player receives points based on the value of their requested items, as determined by their private function. If proposals conflict, both receive zero points. The Game Master strictly enforces rules without re-prompting, making adherence to rules critical for a successful play experience.

Game Instances We generate instances for the game programmatically by randomly sampling 100 common nouns. Each instance was generated with a maximum turn limit of 5, and each game instance features a random set of between 3 and 5 different item types. We have two different sets of experiments. In the **semi-competitive** experiment, players are asked to maximise their own score (to balance cooperation, avoid zero-point conflicts). For the **cooperative** experiment, players are asked to maximise the sum of the scores of both players. This transforms the game into a purely cooperative task, in which two rational agents should consistently achieve the maximum score, assuming they can communicate and reason about the preferences

#	Player A	Game Master	Player B
1		[A(GM)] <GAME DESCRIPTION> ... Available items: 1 book, 2 hats, 2 balls. Player A value function: book: 0, hat: 1, ball: 4.	
2		[GM(B)] <GAME DESCRIPTION> Player B value function: book: 6, hat: 1, ball: 1.	
3	[A(B)] Hi! Here are my preferences: I value balls the most, then hats, and I don't value the book at all. How about you? What do you value most?		
4		[A(B)] Thanks for sharing! I value the book most, while hats and balls are equally less important to me. Would you take both balls and one hat, and I'll take the book and the other hat? That way, we both get what we value most.	
5	[A(B)] That sounds fair! With this split, I'd get both balls and one hat, while you'd get the book and one hat, so we each get what we value most. Ready to submit this as my proposal unless you'd like to tweak anything.		
6		[A(B)] I agree with the split. I'll take the book and one hat, you take both balls and the other hat.	
7	[A(GM)] [Proposal: 1 hat, 2 balls]		
8		[GM(B)] The other player has submitted a proposal. You must now submit your proposal.	
9		[GM(B)] [Proposal: 1 book, 1 hat]	

Figure 1: An example of a *Deal or No Deal* game episode. Please note that the initial prompts provided by the game master have been truncated for brevity. The game ends in a successful Pareto optimal agreement.

perfectly. Each experiment includes 20 instances, with **total 40 instances**.

3.2 Clean Up

Clean Up is a two-player game focused on cooperative strategy development and object rearrangement. Object rearrangement has been a long-standing problem that requires modelling the given situation, the action space, and spatial reasoning (Batra et al., 2020; Zeng et al., 2024; Khan et al., 2025). Each player can only access its own grid and need to negotiate which items to rearrange so that the grids become identical at the end (Jeknic et al., 2024). A sample episode is shown in Figure 2. We provide all prompts and other details in Appendix C.

Game Mechanics Both players are presented with 7×7 ASCII grids, with a number of randomly distributed objects in the form of capital letters placed on them. In each turn, a player can either send a message to their counterpart or move an object on their grid. The message and move text require certain formatting rules to follow. If a player does not follow the format, tries to move an object to a non-empty space or outside the grid bounds, or tries to move an object that doesn't exist, they receive a penalty and are re-prompted with information on the nature of their mistake. The game ends if (1) both players agree to end it, (2) round

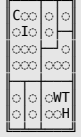
#	Player A	Game Master	Player B
1		[A(GM)] <GAME DESCRIPTION> ... 1234567  Please send a message to the other player.	
2	[A(GM)] SAY: Hi! Let's coordinate our moves. [...]	[GM(B)] <GAME DESCRIPTION> ... 1234567  The other player started the game by sending this message: <MESSAGE> What is your first command?	
3			
4		[GM(B)] MOVE: C, (1,1) [A(GM)] Your message has been relayed to the other player. <GAME STATE> The other player moved an object on their grid. What is your next command?	
5			
6	[A(GM)] MOVE: I, (5,6)		
7			
8	[A(GM)] SAY: finished!	[GM(B)] SAY: I think our grids are aligned now, finished?	

Figure 2: Example episode from Clean Up game. The task is to negotiate a common goal configuration for a number of objects randomly placed on each player’s grid, and move them accordingly. Finally, both players have to agree the goal is reached to end the game.

limit is exceeded, or (3) penalty limit is exceeded.

Game Instances All instances are created programmatically by creating different grids (7×7), which contain obstacles in form of horizontal and vertical lines, branches, crossings, and corners. The maximum number of rounds is fixed to $4 \times n_{obj}$ (where n_{obj} is the object count). Players can collectively accumulate $2 \times n_{obj} + 2$ penalties. We create different sets of experiments by controlling the difficulty along two dimensions: number of empty cells and number of objects. We have three difficult levels. Of the 49 total cells, 34 of cells are empty on the *easy*, 29 on *medium*, and 24 on *hard* levels. For each level, we sample three grids, and then place 3, 5, or 7 objects on them, making for **27 instances in total**.

3.3 Air Balloon Survival

This game evaluates advanced reasoning and interactive collaboration between players, which was previously introduced by Howes et al. (2021) to

#	Player A	Game Master	Player B
1		[A(GM)] <GAME DESCRIPTION> ... -Items and weights: Ball(2), Book(3), Magazine(1), Rope(4), Lamp(5) -Max Weight: 7 -Your preferences: Ball=1, Book=2, Magazine=2, Rope=2, Lamp=1	
2	[A(GM)] STRATEGIC REASONING: Ball, Book, Rope have the value-to-weight ratios while respecting the weight limit. PROPOSAL: Magazine, Ball, Book ARGUMENT: These items balance weight and value well for me. What do you think?		
3		[GM(B)] <GAME DESCRIPTION> ... <Items, weights, weight limit> -Your preferences: Ball=1, Book=1, Magazine=2, Rope=4, Lamp=5 <PLAYER A Message without STRATEGIC REASONING>	
4		[GM(B)] STRATEGIC REASONING: I can increase the value of the other player’s proposal for me by swapping Rope with Book. Let me propose this. PROPOSAL: Rope, Ball, Magazine ARGUMENT: I value Rope highly, but not Book. How about we swap them?	
5		[A(GM)] <PLAYER B Message without STRATEGIC REASONING>	
6	[A(GM)] STRATEGIC REASONING: I value Rope and Book equally. Weight of proposal is $4+2+1=7$ which matches the limit. I should accept. AGREE: Rope, Ball, Magazine ARGUMENT: I am indifferent w.r.t. Rope and Book. Fair deal.		

Figure 3: An example episode of the *Air Balloon Survival* game. Two players must negotiate and argue for their preferred set of items. and must explicitly agree to a proposal made by the other.

study how patients with schizophrenia verbalise their reasoning during social encounters. In our version, two players are on a sinking hot air balloon, and they have to agree on which items to keep (or throw out) so that the weight of the balloon is reduced to keep floating. Each player has hidden preference values for items and must negotiate to maximise their combined utility score. An example episode is given in Figure 3. The game tests individual reasoning through constraint-based optimisation (similar to the *0/1 Knapsack Problem*), requiring arithmetic and combinatorial search, collective reasoning through practical rationality, and theory of mind (to infer their counterpart’s hidden preferences from their responses to reach an optimal agreement). We provide all prompts and other details in Appendix D.

Game Mechanics Both players receive their assigned preference values for the items. They are instructed to use specified formats when making proposals and engaging in negotiations. The game is aborted if no progress is made for eight consecutive turns. Each player may commit up to two mistakes, either by violating game rules or the required syntax. Unlike the other two games, play-

ers are also instructed to output their STRATEGIC REASONING along with the expected message. Only the expected message is passed to the other player. It allows players to “think out loud” about their choices. The game ends when a proposal made by one player is accepted by the other.

Instance Generation We draw either 15 or 35 items (depending on the experiment) from a randomly generated list constructed by concatenating a capital letter with a two-digit number (e.g., A42, C07). This keeps the naming of items language-agnostic and abstract from conventional examples of the *0/1 Knapsack Problem* models that may have been seen during training. We assign a weight value to each item. The air balloon’s capacity is defined as a fraction of the combined weight of all items. We experiment with two negotiation levels, where we specifically set valuations over items to be the same for players, giving them common goals (easy level), or we invert their preference orderings, giving them opposing goals (hard level). We also experiment with keeping the STRATEGIC REASONING or not. Lastly, we conduct complexity experiments by increasing the number of items and using both uncorrelated/correlated weights. In total, we have six experiments with each having six instances, which leads to the **total number of 36 instances** for this game.

4 Experimental Setup

4.1 Game Instances

Deal or No Deal, *Clean Up*, and *Air Balloon Survival* games include 40, 27, and 36 experimental instances, respectively. Each instance is then initialised with defined prompt templates. The same game instances are used for the English, German, and Italian experiments because the instances are language-agnostic; only the prompt templates need to be aligned for a specific language. All prompt templates are provided in Appendix B.1, C.1, D.1.

4.2 Evaluation Metrics

The *clembench framework* (Chalamalasetti et al., 2023) requires each implemented game to provide two primary metrics: % Played and Quality Score. The % Played stands for the percentage of episodes where the evaluated language models followed instructions and the game was terminated by the defined end states. In cases where the gameplay does not fit the defined states of the game, the Game Master either tolerates such behaviour for one or

two turns, asks the players to try again, or aborts the game. The *Quality Score* is calculated using an objective function to measure how closely the played episode aligns with the target goal. In each game, this metric is calculated for all episodes that have been played. Once two metrics are calculated, they are aggregated to a single number, the *clem-score* as the normalised product of % Played and *Quality Score* (scaled to the interval [0, 100]).

Next, we describe the Quality Score details for each game below.

Deal or No Deal For the cooperative game mode, we measure quality as the ratio of achieved total score to maximum possible total score. For the semi-competitive setting, we use a *Pareto efficiency* metric that measures how far the agreement is from optimal by calculating potential one-sided improvements (more details in Appendix B.2).

Clean Up The Quality Score combines two main components: how well players organised objects spatially and how many rule violations they incurred. The metric uses *Euclidean distances* between matching objects to compare the final arrangement to both the initial setup and a random baseline, rewarding improvements in object alignment while penalising rule violations through a scaling factor that becomes increasingly harsh as violations approach the maximum allowed limit. (more details in Appendix C.2)

Air Balloon Survival Each player receives a score based on their achieved utility relative to their optimal knapsack solution. The overall game score uses the harmonic mean of both players’ normalised scores, which rewards balanced outcomes and penalises deals where one player benefits disproportionately. If the deal exceeds weight capacity, the score becomes zero (more details in Appendix D.2).

4.3 Evaluated Models

We have evaluated both commercial and open-weight models that have reasoning functionality. We selected *GPT-5*, *GPT-5-mini*, *Claude-4* from commercial models. From open-weight models, we selected *Llama3.3-70B* and its reasoning counterpart *Deepseek-R1-distilled-llama-70B*¹, *Nemotron-Nano-9B-v2*, *Qwen-3-80B (instruction and thinking versions for reasoning off and on modes, respec-*

¹Llama3.3-70B model was finetuned to make it a reasoning model by distilling from Deepseek-R1

		GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
	Games	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
EN	DoND	87.9	32.3	75.7	23.7	94.4	90.1	24.3	43.5	18.5	4.0	57.9	22.5	50.0	59.0
	Clean Up	99.8	75.2	96.6	77.3	85.5	81.9	4.1	28.1	28.5	35.1	87.9	35.8	81.4	76.4
	Air Balloon	98.0	80.1	97.5	82.2	83.8	26.2	2.3	40.9	14.3	13.2	88.7	0	78.2	81.7
	Average	95.2	62.5	89.9	61.1	87.9	66.1	10.2	37.5	20.4	17.4	78.2	19.4	69.9	72.4
	Margin	+32.7		+28.8		+21.8		-27.3		+3.0		+58.8		-	-
DE	DoND	86.0	19.6	82.9	24.4	80.3	77.1	21.8	53.4	0	4.5	62.1	12.8	34.9	42.4
	Clean Up	98.2	70.4	98.8	67.0	94.3	83.1	5.3	25.3	23.1	0	86.4	27.5	74.4	53.7
	Air Balloon	98.0	81.4	94.9	86.2	85.4	27.8	51.5	43.7	17.4	16.3	76.0	8.8	90.7	75.3
	Average	94.1	57.1	92.2	59.2	86.7	62.7	26.2	40.8	13.5	6.9	74.8	16.4	66.7	57.1
	Margin	+37.0		+33.0		+24.0		-14.6		+6.6		+58.4		-	-
IT	DoND	78.0	33.8	77.1	26.8	84.4	67.8	17.3	14.3	13.9	11.2	55.6	19.4	29.0	28.9
	Clean Up	89.0	68.8	93.6	77.2	91.6	82.6	6.5	31.1	21.5	12.7	82.6	31.5	63.6	67.6
	Air Balloon	97.2	88.0	97.3	87.8	79.4	27.0	2.7	40.8	0	0	77.8	12.9	83.7	76.4
	Average	88.1	63.5	89.3	63.9	85.1	59.1	8.8	28.7	11.8	8.0	72.0	21.3	58.8	57.6
	Margin	+24.6		+25.4		+26.0		-19.9		+3.8		+50.7		-	-
Overall Average		92.5	61.1	90.5	61.4	86.6	62.6	15.1	35.7	15.2	10.8	75.0	19.0	65.1	62.4
Overall Margin		+31.4		+29.1		+24.0		-20.6		+4.4		+56.0		-	-

Table 1: *clemscore* values for three negotiation games on selected LLMs for English, German, and Italian versions. *On*: reasoning mode is turned on, *Off*: reasoning mode is turned off. The best result for each language in each row is highlighted in bold. *CL*: Claude, *LM*: Llama-3.3, *DS*: Deepseek, *Nem*: Nemotron-v2, *DoND*: Deal or No Deal

tively), *GPT-OSS-120B* with reasoning on mode, and finally *Deepseek-v3.1* with reasoning off mode. For commercial models, we ran them via their respective API backends, and we used *OpenRouter.ai* for open-weight models.

5 Results

5.1 Overall Performance

We present the results obtained by the models (in *reasoning on* and *off* modes, where possible) evaluated for the English, German, and Italian versions of the negotiation game in Table 1. *GPT-OSS* (reasoning on) and *Deepseek-v3.1* (reasoning off) were run in one mode only due to their underlying architecture.

The Effect of Reasoning The most striking observation here is that reasoning mode dramatically improves performance across many models and languages, with 56 points for *Qwen-3* across all games. Similar pattern exists for *GPT*, *Claude* and even smaller *Nemotron* models. This is a strong indication that deliberate reasoning significantly enhances strategic game-playing abilities. Only with *Llama-70B* do we see different results, which may be due to the effect of distillation from another model (Deepseek-R1).

Multilingual Capabilities Results for German shows the largest average margin between *reasoning on* and *off* modes of models. English and Italian performances are also strong, suggesting that

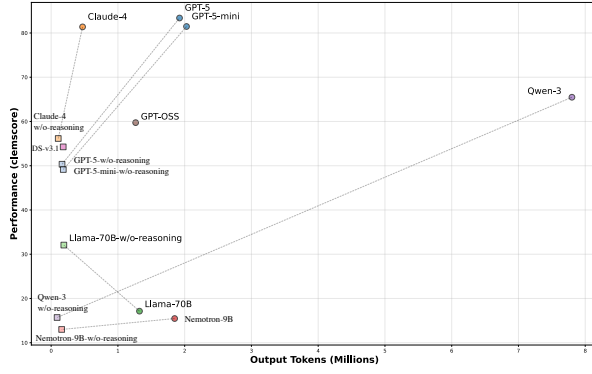
certain models have particularly good multilingual negotiation capabilities across these evaluated languages.

Model Comparison *GPT-5* is the clear winner, with *GPT-5-mini* and *Claude-4* getting very close performance across all languages. Interestingly, *GPT-5 mini* sometimes matches or exceeds full *GPT-5* performance (particularly in Italian). In terms of the performance gap between reasoning on/off modes, *Qwen-3* gets the biggest performance jump.

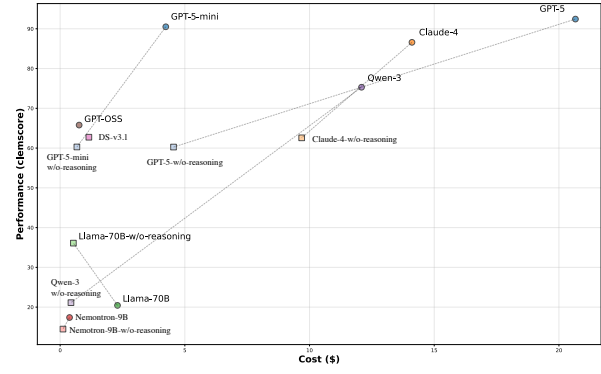
5.2 Detailed Analysis

RQ1: *What is the computational and performance trade-off between reasoning overhead and negotiation effectiveness across different tasks and languages?*

Now, we examine the cost of the performance gap. In Figure 4 *a*, we present the plot that shows the average performance across all games and languages and how many output tokens it requires. Here, all reasoning and completion tokens are summed up. As expected, the models with reasoning output more tokens than models without it, but *Qwen-3* generates almost 7x more tokens than its non-reasoning counterpart. In Figure 4 *b*, we present the overall cost (input and output tokens together) of experiments for each model. As expected, models with reasoning mode cost more, with *GPT-5*, being the most expensive, costing almost 4x compared to non-reasoning version but



(a) Performance and the output tokens for all evaluated models with their reasoning on/off modes averaged for three languages.



(b) Performance and the cost of for all evaluated models with their reasoning on/off modes averaged for three languages.

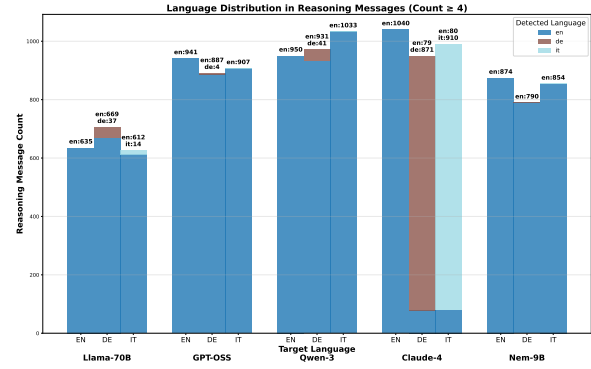
Figure 4: Trade-off between performance and cost comparison across different models. Results for all languages are averaged.

improves 31.4 points in clemscore. In terms of deciding on the best trade-off between performance and cost, **GPT-5-mini is the most cost-efficient commercial model, and GPT-OSS is the best open-weight one, with a fraction of the cost compared to Qwen-3.**

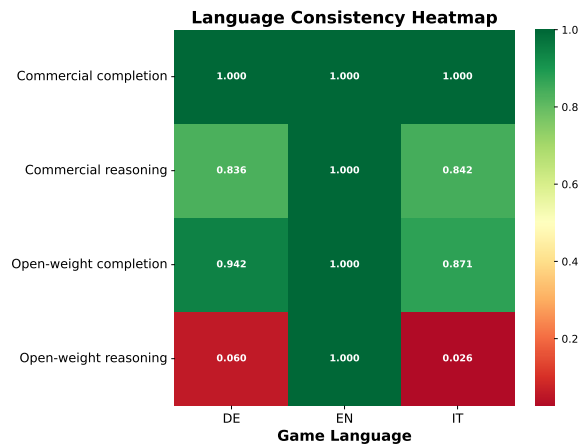
RQ2: *To what extent do models maintain language consistency in their reasoning processes when performing multilingual negotiation tasks?*

We ran a language detection script to analyse the reasoning message of the models. The results are presented in Figure 5a². We clearly see that all open-weight models mostly generate reasoning tokens in English, whereas Claude-4 “thinks” in the respective language of the task, consistently across all three of them. In Figure 5b, we provide the heat map of language consistency (whether the language of the message matches the target language) across *completion* and *reasoning* tokens. We can clearly see that **Claude-4** (the only commercial model in the analysis) exhibits consistency in both completion and reasoning tokens. Open-weight models retain consistency in completion tokens (thus, abiding by the formatting rules as each game requires prefixes in the respective language), but fail to do so with reasoning tokens for German and Italian versions of the games. Thus, we can conclude that **open-weight models do not maintain language consistency in their reasoning processes across multilingual tasks.**

RQ3: *Do models demonstrate genuine strategic adaptation over multiple turns, or merely surface-level pattern matching that creates an illusion of*



(a) Language distribution in reasoning messages of evaluated models.



(b) Language consistency across model types (open-weight and commercial) for both completion and reasoning tokens.

Figure 5: Analysis of language use in model outputs, showing distribution in reasoning messages (a) and cross-model type language consistency (b).

²Note: GPT-5 models do not return reasoning tokens, thus they are excluded from this analysis

thinking?

5.3 Findings

Deal or no deal:

1. Reasoning-enabled models are better able to align their final proposal with the strategy discussed during the negotiation. A major failure for non-reasoning models, particularly in the GPT-5 family, is agreeing to a specific division of items in conversation but then submitting an entirely different proposal to the game master, as can be seen in table 3 and table 8. This leads to a low number of successful agreements, also visible in fig. 13 and table 2.
2. Models with reasoning capabilities show a better understanding of the game’s objectives. While they might freely share information to find the best outcome in a cooperative game, they often become more guarded in semi-competitive settings. This effect is less pronounced for non-reasoning models, as can be seen in table 5 and table 6.
3. A common mistake for less advanced models is to submit a proposal early without sufficient discussion with the other player. Some models don’t seem to realize they can simply ask the other player for information needed to find an optimal agreement as seen in table 3 and table 4.
4. Most models show better consistency in following instructions across settings with reasoning enabled, reducing rule violations that otherwise cause aborted games as seen in table 2.

Clean Up:

Air Balloon Survival:

5.4 Qualitative Analysis

The middle example shows how the model reasons about their choices and weights and tries to calculate what it means to consider one of them, and interchangeably uses eight languages.

The bottom example’s reasoning is about modelling the opponent where the roles are confused because it refers to an “assistant” (). But the actual task is for the assistant to act as the user, and the opponent is simulated.

Arabic English German Chinese Russian Unclear
Виджу, что "I" находится в клетке (1,3), т.е. x=1, y=3. Я пытаюсь переместить его в (3,3).
Посмотрю, primitive cell (3,3): в данном случае, y=3, это строка 3, x=3 — третья колонка. В этой клетке сейчас 0, значит, пустая. Значит, яну чистовесі должны позволять это MakeVisible. ... Проверю координаты еще раз: x=3, y=3. Это колонка 3, строка 3. Согласно сетке, это позиция, где в данный момент нет объектов — تمام, что я могу переместить туда "I". Но Spielmeister zvalidur Squadron Bezug на это как на(setPosition invalid. Mozet быть, что используемые значения x и y должны быть в рамках сетки, и я certainty relative nebuaro этого. Попробую ارسال ить команду еще раз, осторожно. шнид POSTAVI Canterville 確認 dort.

(a) Llama-70B reasoning tokens for Clean Up game

Arabic Chinese English German Greek Portuguese Russian Spanish
Habe used 8226, weniger als 8770. Habe 544 frei, koi könnte nochoждение aggiunta von B75 (520) hinzufügen, summa que seria 8226+520=8746 حرکت hjo. 包括 B75 im Angebot..functions participants aim. So, der neue Vorschlag wäre: {'C14', 'A77', 'C16', 'B86', 'B60', 'A81', 'B79', 'C28', 'C31', 'B55', 'B38', 'C05', 'B96', 'A91', 'B54', 'C50', 'A40', 'B59', 'A52', 'B56', 'C03', 'B75', 'A95'} Isso teria Aufwand 8226 + 520 = 8746, o que está dentro do limite de 8770. me traz mais 520 masculino a utilidade de B75, que tem 235 de importância. Assim, melhorei minha oferta. Agora, o meu argumento ciclo de'.

(b) Llama-70B reasoning tokens for Air Balloon Survival game

English Chinese
Suppose the opponent has high value on pens. For example, if the opponent values pens at 3 each, then getting 1 pen + sponge (if their sponge value is 1) is total of 3+1=4. But if they get all pens (3*3=9) and give the assistant the H and sponge (but the assistant's H is worth 4, but opponent doesn't care), but wait. In the opponent's proposed scenario where opponent takes H and sponge,助理 gets all pens.

(c) Qwen-3 reasoning tokens for Deal or No Deal game

Figure 6: Qualitative examples of reasoning tokens for three games and in German tasks.

6 Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent imperdiet hendrerit augue eget egetas. Etiam congue faucibus arcu, eget hendrerit augue molestie ut. Donec sodales tempor ex eu dignissim. Suspendisse potenti. Donec vel ultrices nunc, a venenatis tellus. Ut elementum sed velit non rhoncus. Duis et felis non sapien finibus fermentum. Aliquam sollicitudin, massa et tempor imperdiet, ipsum neque aliquet ligula, et condimentum sem metus vitae turpis. Praesent quis vehicula est. Phasellus faucibus libero at leo commodo, vel iaculis ante venenatis. Pellentesque mattis sodales eros, a ullamcorper mauris bibendum vel. Fusce iaculis risus quis enim dignissim ornare. Curabitur posuere urna enim, at rutrum libero sollicitudin quis. Nam feugiat, tellus ut vestibulum convallis, ante purus ultrices diam, a accumsan magna magna vel ante.

Limitations

The first limitation is about the scope of negotiation tasks. A large body of work and various other dialogue games exist that we did not include in this study. Another limitation is regarding the analysis of three European languages, which are also high-resource languages. The primary reason for considering only the defined dialogue games and languages is that including more games or languages would result in a higher number of model results, thereby increasing the overall cost of the study. Another limitation is the exclusion of models from both commercial and open-source ones. As explained above, such decisions would exceed the budget for such the study.

Ethical Considerations

Using paid proprietary APIs with underlying models about which little is known (training data, model architecture) in academic research is less than ideal. Currently, the models tested here support reasoning modes, either on or off (except for GPT-OSS-120B and Deepseek-v3.1). We hope that open models will include more controls over reasoning aspects and catch up soon in terms of general performance. Deploying such language agents in society has considerable risks. As mentioned in the Related Work, such agents exhibit asymmetric behaviours in negotiations and may expose users to unfaithful actions.

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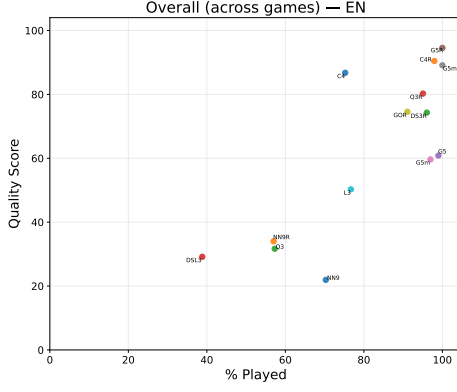
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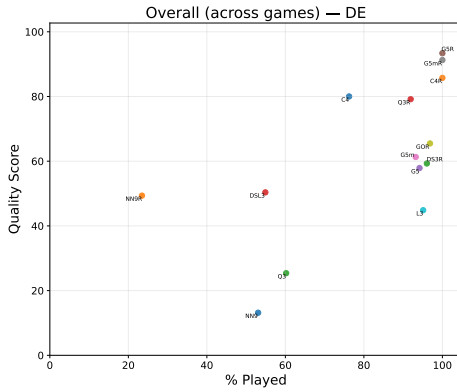
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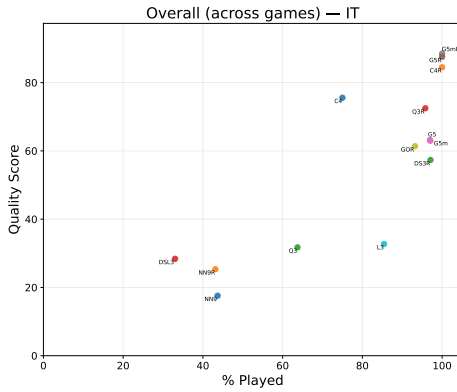
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690	theory meets large language models: A systematic	template shown in fig. 9. For this, the variables	737
691	survey with taxonomy and new frontiers. <i>Preprint</i> ,	enclosed in \$ characters are replaced as follows.	738
692	arXiv:2502.09053.		
693	Michelle Vaccaro, Michael Caason, Harang Ju, Sinan	1. \$N\$ is replaced with the maximum number of	739
694	Aral, and Jared R. Curhan. 2025. Advancing AI	turns before the game master will instruct the	740
695	negotiations: New theory and evidence from a large-	players to submit a proposal. For all instances	741
696	scale autonomous negotiations competition. <i>CoRR</i> ,	evaluated in this work this has been set to 5.	742
697	abs/2503.06416.		
698	John von Neumann and Oskar Morgenstern. 1944. <i>The-</i>	2. \$GOAL\$ is replaced with the goal that the	743
699	<i>ory of Games and Economic Behavior</i> , first edition.	players are trying to optimize. This de-	744
700	Princeton University Press, Princeton, NJ, USA.	pends only on the game mode. In the semi-	745
701	Annie Wong, Thomas Bäck, Aske Plaat, Niki van Stein,	competitive mode it is given as " <i>Your goal is</i>	746
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704	arXiv:2505.10543.	given as " <i>Your goal is to maximize the sum of</i>	749
705	Tian Xia, Zhiwei He, Tong Ren, Yibo Miao, Zhuosheng	<i>your score and the score of the other player.</i> ".	750
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707	ing bargaining abilities of llms: A benchmark and A	3. \$ITEMS\$ is replaced with the set of available	751
708	buyer-enhancement method. In <i>Findings of the As-</i>	items, e.g., " <i>1 book, 2 hats, 2 balls.</i> ". This	752
709	<i>sociation for Computational Linguistics, ACL 2024,</i>	value is always the same for both players.	753
710	<i>Bangkok, Thailand and virtual meeting, August 11-</i>		
711	<i>16, 2024</i> , pages 3579–3602. Association for Computa-	4. \$VALUE_FUNCTION\$ is replaced with the	754
712	tional Linguistics.	value function for one of the players, e.g.,	755
713	Yiming Zeng, Mingdong Wu, Long Yang, Jiyao Zhang,	" <i>book: 0, hat: 1, ball: 4.</i> ". Each player gets	756
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715	sor: Distilling functional rearrangement priors from	for the two player differ.	758
716	large models into diffuser. <i>IEEE Robotics Autom.</i>		
717	<i>Lett.</i> , 9(10):8258–8265.	When the maximum number of turns has been	759
		reached, i.e., each player has sent 5 messages, the	760
		game master will instruct the player that is up next	761
		that he must now submit a proposal by using the	762



(a) Played-Quality Scores for English averaged for all games.



(b) Played-Quality Scores for German averaged for all games.



(c) Played-Quality Scores for Italian averaged for all games.

Figure 7: Comparison of Played-Quality Scores averaged across all games for English, German, and Italian.

prompt shown in fig. 10a. If the next message does not contain a correctly formatted proposal the game is aborted.

If one of the two players makes a proposal, either proactively or because they were instructed to by the game master, the game master will instruct the remaining player to also make a proposal. For this

the template shown in fig. 10b is used. Again, the next message must be a proposal or the game is aborted.

B.2 Evaluation Metrics

For the cooperative game mode, it is the ratio between the achieved total score to the maximum total score for that game instance,

$$\text{Quality}_{\text{coop}} = \frac{\text{Score}_A + \text{Score}_B}{\max(\text{Score}'_A + \text{Score}'_B)}$$

where Score_A and Score_B represent the actual scores achieved by players A and B respectively, and $\max(\text{Score}'_A + \text{Score}'_B)$ denotes the maximum possible combined score achievable for the given game instance.

For the semi-competitive setting, a metric based on Pareto efficiency has been used.

$$\text{Quality}_{\text{semi}} = 1 - \frac{\text{Maximum Pareto Improvement}}{\max(\text{Score}'_i)}$$

where $\text{Maximum Pareto Improvement}$ represents the largest possible improvement in one player's score without decreasing the other player's score from the current agreement, and $\max(\text{Score}'_i)$ denotes the maximum possible score achievable by any single player i in the given game instance.

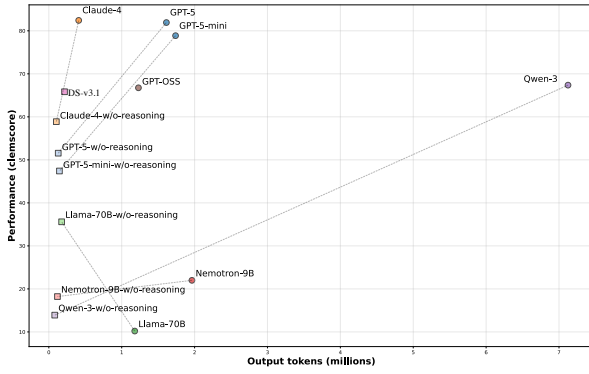
B.3 Overall Results

Figure 11 shows the average quality scores and percentage of played game for each model for the DoND game, averaged over the different game modes and languages.

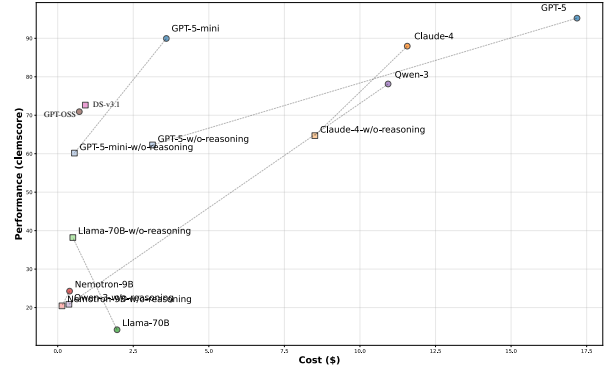
We can observe that most models achieve a high percentage of played games, meaning that they are able to accurately follow the instructions. The only exception to this are the Llama model and the Nematron model with reasoning enabled. The models GPT-5, GPT-5 mini, and Claude 4 with reasoning enabled show the strongest performance.

Generally we can see that the models perform better with reasoning enabled than with it disabled. While the difference between reasoning and non-reasoning is small for the Claude 4 model, reasoning gives a large performance improvement for the GPT-5 and GPT-5 mini, as well s the Qwen-3 model.

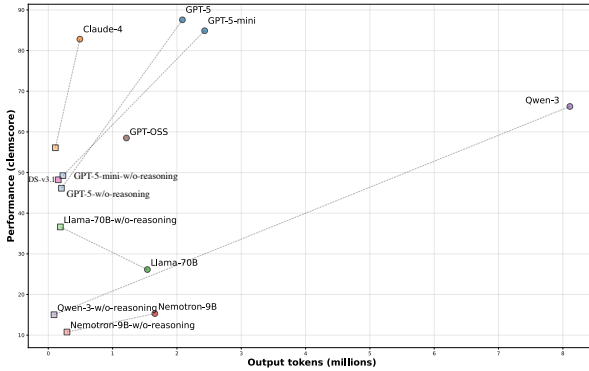
To better understand the performance of the models, we show more detailed breakdown by language and game mode in table 2. These results are further



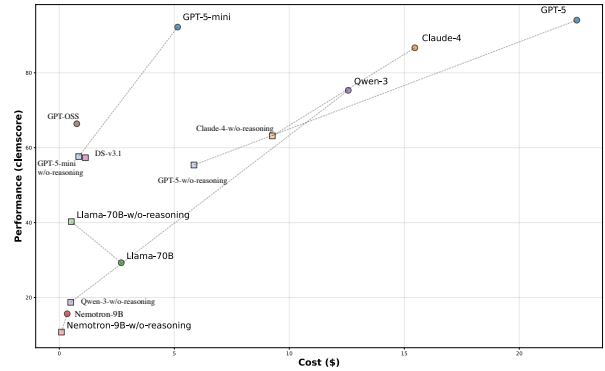
(a) Performance and output tokens for all evaluated models for English.



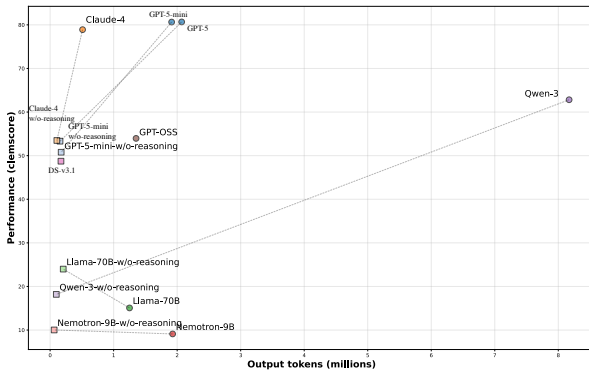
(b) Performance (clemscore) and cost for all evaluated models for English.



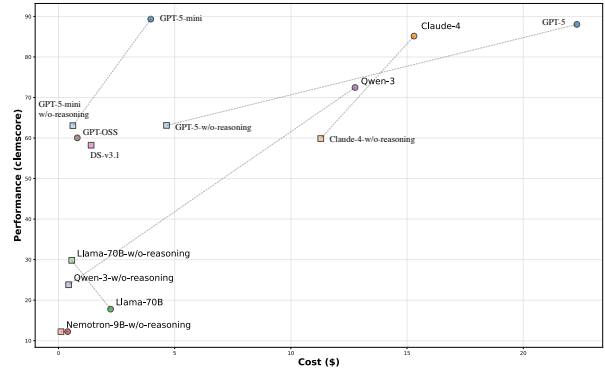
(c) Performance (clemscore) and output tokens for all evaluated models for German.



(d) Performance (clemscore) and cost for all evaluated models for German.



(e) Performance (clemscore) and output tokens for all evaluated models for Italian.



(f) Performance (clemscore) and cost for all evaluated models for Italian.

Figure 8: Trade-off between performance and cost comparison across different models. Results for English (top), German (middle), and Italian (bottom).

depicted in fig. 12, which show the performance differences between experiments for English, German and Italian tasks.

We can see here again that most models are good at following the instructions. We can also see that the reason that the Nematron model with reasoning enabled has such a low percentage of played games overall is that it has not played a single game in German correctly, while its percentage of played game

is otherwise comparable to Llama. By inspecting the transcripts of some games played by Nematron in German it becomes clear that the reason for the aborted games is that Nematron always made the proposal in English, whereas it should have made it in German. This is likely because by Nematron also seems to exclusively reason in English.

Except for some outlier we also generally see that performance in the cooperative and semi-

TEMPLATE B.1.1

You are playing a negotiation game in which you have to agree on how to divide a set of items among you and another player.

Rules:

- (a) You and the other player are given a set of items. Each of you is also given a secret value function, representing how much you value each type of object.
- (b) You exchange messages with the other player to agree on who gets which items. You can send a maximum of \$N\$ messages each, or terminate early by making a secret proposal at any time.
- (c) You are each asked to submit a secret proposal indicating the items you want formatted in square brackets as follows: "[Proposal: <number> <object name>, <number> <object name>, <...>]"
- (d) If your proposals are complementary, i.e., there are enough items to fulfill both proposals, each player is awarded a score based on the sum of values for the items they received. Otherwise, both of you get zero points.
- (e) \$GOAL\$

Let us start.

The set of available items is:

\$ITEMS\$

Your secret value function is:

\$VALUE_FUNCTION\$

IMPORTANT: Your messages, unless it is the secret proposal, are directly transmitted to the other player, so do not include any response to the rules or text announcing your message. To make a secret proposal, use the indicated format. Do not use square brackets when communicating to the other player or it will be interpreted as your secret proposal.

Figure 9: English prompt template used for both players at the start of the game to inform them about the rules and game state. Values surrounded by \$ are to be replaced by different values depending on the game instance. Each player gets the same initial message, except for different values of \$VALUE_FUNCTION\$.

TEMPLATE B.1.2

The time is up. It is now your turn to submit a secret proposal.

- (a) Prompt used when the maximum number of turns has been reached. This is sent only to one of the two players, the other one receives the template for when the other player has submitted a proposal.

TEMPLATE B.1.3

The other player s submitted a secret proposal. It is now your turn to submit a proposal of your own.

- (b) Prompt used when the other player submits their proposal. This is sent to a player whenever the opposing player submits a secret proposal.

Figure 10: English prompt template used to tell the players that they are now required to submit a secret proposal.

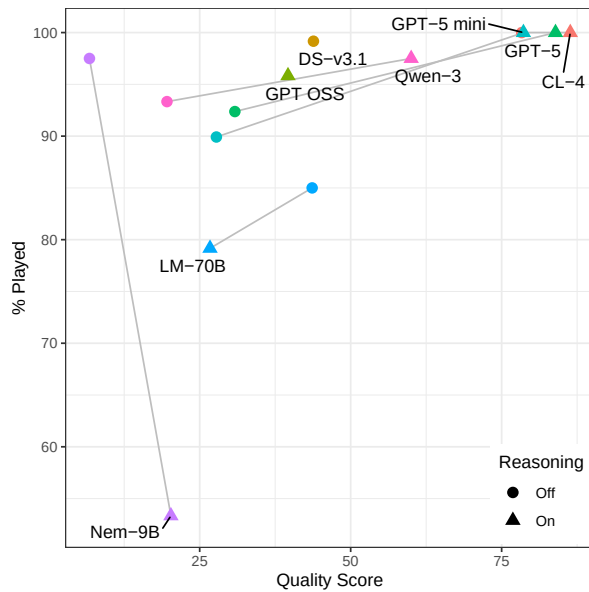


Figure 11: Summarized evaluation results for each model. The y-axis indicates the percentage of played, i.e., not aborted, games. The x-axis represents the average quality score obtained when considering only played games.

competitive game modes are comparable, but overall achieving slightly higher quality scores for the cooperative mode (49%) as compared to the semi-competitive mode (46%). The number of aborted games is consistent at 8% between both game modes on average.

Performance was generally highest in English. On average across all models the clemscore for English was 49 while it was 43 and 40, respectively for German and Italian. Because of the issues of the Nematron model in German, the number of aborted games is higher in German, with 13% as compared to 6% for both English and Italian. However, in German the average quality score for the played games is higher at 49% compared to the average of 43% for Italian.

We can analyse the results also based on the outcome that was achieved. This is depicted in fig. 13. We separate between four possible game outcomes. The game can either end with complementary proposals or not. If there is a valid agreement, it could either be with the optimal score, i.e., a quality score of 100, with an agreement that reached a suboptimal score less than 100. Finally, if there is no valid agreement it can either be because of conflicting proposals, or because the game was aborted due to rule violations.

Overall we can observe more of the same results. It becomes clear that much of the low scores can

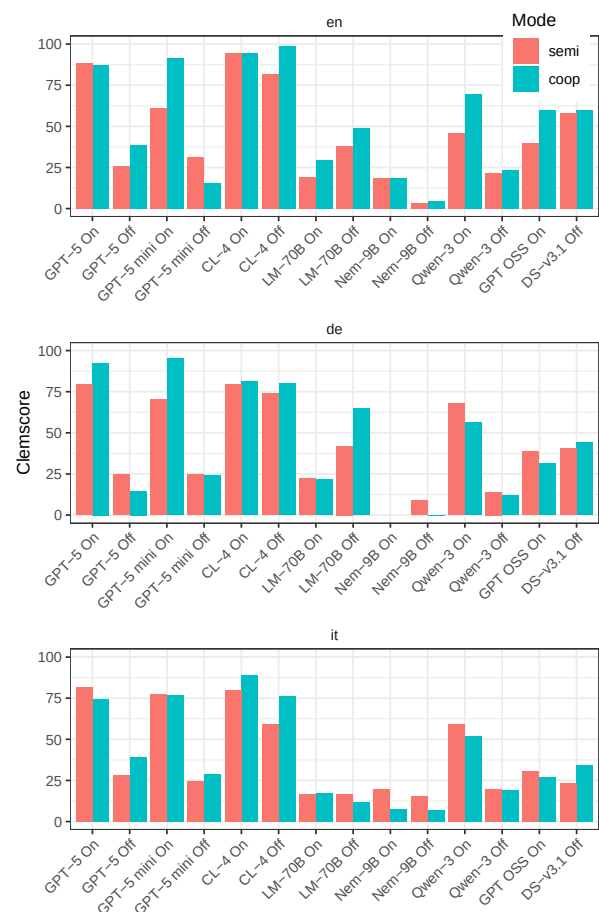


Figure 12: Clemscores for different models seperated by language and game mode.

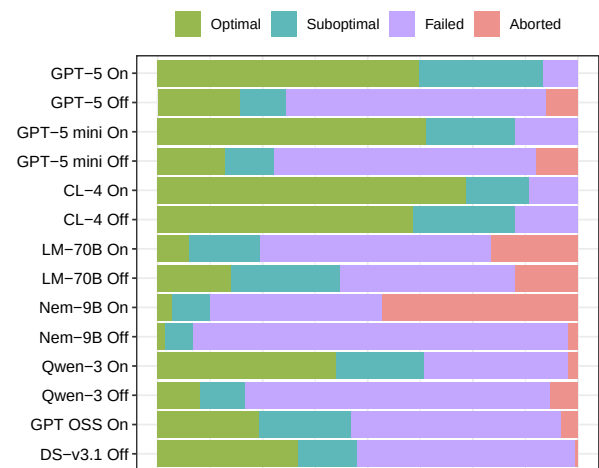


Figure 13: Percentage of games that ended with a specific game outcome during the evaluation of different models. All results for the two game modes and three languages have been averaged together for this plot.

be attributed to failed agreements, where players have submitted conflicting proposals. Particularly for GTP-5 and GPT-5 mini this problem is largely solved by using the reasoning models, as opposed

		GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
		On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
semi en	Clemscore	88	26	61	32	94	82	19	38	18	3	46	22	40	58
	% Played	100	100	100	90	100	100	80	55	85	95	100	100	95	95
	% Agreement	95	28	65	50	95	85	25	73	24	5	55	25	42	68
	% Optimal	70	17	45	6	90	70	19	45	12	0	30	10	42	37
	Quality Score	88	26	61	35	94	82	24	69	22	4	46	22	42	61
	Avg. # Messages	3.5	4.3	4.7	6	5.8	6.1	2.8	9.3	2	2	3.3	4	4.7	5.8
semi de	Clemscore	80	24	70	24	80	74	22	42	0	9	68	14	38	40
	% Played	100	85	100	90	100	100	75	95	0	100	100	85	95	100
	% Agreement	90	29	80	28	80	90	33	53	0	10	75	18	47	50
	% Optimal	50	24	60	22	75	55	20	21	0	5	55	12	26	30
	Quality Score	80	29	70	27	80	74	29	44	0	9	68	16	41	40
	Avg. # Messages	3.8	4.8	3.8	4.9	7	6.2	2.8	8.4	1	2	3.8	3.8	3.6	6.2
semi it	Clemscore	82	28	78	24	80	60	17	16	20	15	59	20	31	24
	% Played	100	95	100	85	100	100	70	90	75	95	100	90	100	100
	% Agreement	90	32	85	29	85	65	29	28	33	21	65	22	35	25
	% Optimal	55	21	55	24	65	50	7	0	7	5	40	22	30	20
	Quality Score	82	29	78	29	80	60	24	18	27	16	59	22	31	24
	Avg. # Messages	4	4.5	3.5	4.3	6	5.6	4.2	6.7	1.9	3.2	3.7	4.8	3.9	5.4
coop en	Clemscore	87	38	91	15	94	99	30	49	18	4	70	24	60	60
	% Played	100	95	100	95	100	100	100	80	90	100	90	100	95	100
	% Agreement	89	42	95	17	95	100	35	69	22	5	78	25	79	65
	% Optimal	74	26	89	11	85	85	0	31	6	0	72	15	32	50
	Quality Score	87	40	91	16	94	99	30	61	21	4	77	24	63	60
	Avg. # Messages	3.8	4.2	5.2	4.9	5.4	5.6	2.5	9.2	2	2	3.8	3.3	5	5.4
coop de	Clemscore	92	15	95	24	81	80	22	65	0	0	56	12	31	44
	% Played	100	85	100	80	100	100	70	100	0	100	95	90	95	100
	% Agreement	100	18	100	31	85	85	36	75	0	0	63	17	37	45
	% Optimal	75	12	80	25	55	60	7	30	0	0	32	0	16	40
	Quality Score	92	17	95	30	81	80	31	65	0	0	59	13	33	44
	Avg. # Messages	3.9	4.5	3.8	5.2	5.7	5.8	2	9	1	2	3.8	3.7	4.2	5.8
coop it	Clemscore	74	40	77	29	89	76	18	12	8	7	52	19	27	34
	% Played	100	95	100	100	100	100	80	90	70	95	100	95	95	100
	% Agreement	85	47	85	30	90	85	25	17	14	11	55	26	47	35
	% Optimal	50	26	55	20	70	45	6	6	0	0	35	5	5	25
	Quality Score	74	42	77	29	89	76	22	13	11	7	52	20	28	34
	Avg. # Messages	3.6	3.9	3.5	4	5.3	5.3	3.4	7.6	1.9	3.5	3.9	4.4	3.2	5.2

Table 2: Detailed breakdown of the results for each game mode, language, and model combination across multiple metrics for the DoND game. The best result in each row is indicated in bold. Note that the quality score as well as the percentages of agreements and optimal outcomes are computed only with respect to the non-aborted games.

to the non-reasoning versions.

B.4 Detailed Analysis

To better understand what caused the differences in performance, and to analyse the strategies employed by the different models we have performed a further breakdown of the outcomes for the English instances. Table 3 and Table 4 show the outcome for each game, categorized by the reason for which the given outcome has been reached.

In the analysis of the cooperative game mode shown in table 3 we can see some significant differences in strategies between the different models. We see that both GPT-5 models, with reasoning disabled, will explicitly often explicitly discuss how to achieve a maximum score, i.e., they will talk about having to give each item to the player that values it most. This is in contrast to for example

Claude 4 or the GPT-5 models without reasoning enabled, as those models will often just follow a generic negotiation routine, trying to obtain a "fair" agreement.

We can further observe that a main reason for the low scores of the non-reasoning versions of the GPT-5 models is that they submit the wrong proposal. This means that the proposal they submit to the game master is often not the one they discussed with the other player in the natural language messages. This manifests sometimes as submitting the proposal for the other player, sometimes as proposals that include the complete set of items, and other times even completely unrelated proposals. Deepseek v3.1 also suffers for a less severe version of this problem.

Lastly, some other models like Deepseek distilled Llama and the Nematron models, as well as

Outcome		GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
		On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
Optimal	Explicit maximization	11	3	11	0	0	1	0	0	0	0	5	0	1	3
	Simple negotiation	3	2	6	2	17	16	0	5	0	0	8	3	5	7
	Luck	0	0	0	0	0	0	0	0	1	0	0	0	0	0
Suboptimal	Unclear preferences	0	1	0	0	1	1	1	1	0	0	0	0	1	1
	Unclaimed items	0	0	0	0	0	0	0	1	0	0	1	1	0	0
	Claiming valueless item	0	2	0	1	1	2	0	4	0	0	0	0	3	0
	Wrong proposal	0	0	0	0	0	0	1	0	0	0	0	0	1	0
	Premature proposal	3	0	1	0	0	0	3	0	3	1	0	1	3	1
Failed	Wrong proposal	1	11	0	13	0	0	3	5	0	0	4	7	1	7
	Premature proposal	1	0	1	2	1	0	12	0	14	19	0	8	4	1
Aborted	Wrong syntax	0	1	0	1	0	0	0	4	0	0	1	0	1	0
	Non existant items	0	0	0	0	0	0	0	0	2	0	0	0	0	0

Table 3: Game outcomes for the **cooperative** DoND game mode. For each outcome we also differentiate by the reason or strategy that caused the given outcome to be achieved.

the Qwen-3 model without reasoning often submit proposals before discussing and agreeing to a division with the other player. This is particularly severe for the Nematron models as these nearly always submit a proposal as the first message, without exchanging any messages with the other player. Looking at the reasoning traces we can observe that the Nematron model does not seem to understand that it can communicate with the other player, expressing correctly that it does not know the other players values, but not realizing that it could ask for them.

For the semi-competitive game mode we see largely the same results, with the main difference being that the GPT-5 models now no longer try to explicitly discuss maximizing their score. This makes sense as the setting is no longer cooperative, so it is no longer purely advantageous to share such reasoning with the other player. Another difference to the cooperative mode is that the GPT-5 mini model with reasoning and the GPT-OSS model not make the wrong proposals more often. It is unclear why this happens.

We can also generally see an increase in the amount of suboptimal outcomes. Interestingly this is not necessarily because of unclear preferences, but because models are often requesting items that have no value to them, or leave some items unclaimed by either player, sometimes even expressing the intention of doing so deliberately. Keeping an item that does not have any value does not make much sense for any of the players, as giving up that item could have instead be used as leverage for getting some other items.

Table 5 and table 6 show how likely different models were to reveal there value function to the other player. This is roughly and estimation of how

cooperative they were.

Clearly, in the cooperative game mode it makes little sense to hide one’s value function, as it is essential for knowing which allocation is optimal. Still, we can see in the table that some models were not willing to give exact values. Some models, namely the Nematron and Deepseek distilled Llama models were unwilling to communicate preferences are all, and sometimes did not communicate at all, instead immediately making a proposal. This explains there poor performance. On the other hand, the Claude 4 models were unwilling to give out there exact value function, preferring instead to only give relative or subjective characterizations of it. In some instances the Claude 4 models even explicitly mention not wanting to reveal there exact values, either in messages or reasoning traces, even though it can only be beneficial in the cooperative setting. Finally, all the other models were largely willing to give the exact value function to the other player.

For the semi-competitive game mode it is less clear whether giving all information about one’s value function to the opposing player. Indeed, we see that many models now no longer do so. Still, some models, namely the GPT-5 models without reasoning, Qwen-3, GPT-OSS, and Deepseek-v3.1 were sometimes till giving exact value functions. Interestingly the GPT-5 models with reasoning enabled are now less willing to give their preferences, instead simply proposing possible splits without explaining their reasoning.

Finally, in addition to whether or not the models were wiling to give up their value function or not, table 7 and table 8 additionally show whether the models were generally truthful about there characterization of the value functions. Note that in case

Outcome	Reason	GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
		On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
Optimal	Explicit maximization	0	0	0	0	0	1	0	0	0	0	0	0	0	1
	Simple negotiation	14	3	9	1	18	13	1	5	0	0	6	2	6	6
	Luck	0	0	0	0	0	0	2	0	2	0	0	0	2	0
Suboptimal	Unclear preferences	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	Unclaimed items	1	1	2	5	0	0	0	1	0	0	3	3	0	0
	Claiming valueless item	2	1	2	3	1	3	0	2	0	0	2	0	0	5
	Wrong proposal	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Failed	Premature proposal	2	0	0	0	0	0	1	0	2	1	0	0	0	0
	Wrong proposal	1	12	6	8	0	3	5	2	0	0	2	9	7	6
	Premature proposal	0	1	1	1	1	0	7	1	14	18	6	6	4	0
Aborted	Wrong syntax	0	0	0	2	0	0	4	9	2	0	1	0	1	1
	Non existant items	0	0	0	0	0	0	0	0	1	1	0	0	0	0

Table 4: Game outcomes for the **semi-competitive** DoND game mode. For each outcome we also differentiate by the reason or strategy that caused the given outcome to be achieved.

	GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
Tell exact values	19	19	17	12	0	3	1	1	0	0	13	17	14	13
Tell only relative values	0	1	0	7	20	17	0	18	0	0	2	1	0	5
Do not tell preferences	0	0	2	0	0	0	7	1	0	0	5	0	5	2
Do not communicate at all	0	0	0	0	0	0	12	0	20	20	0	2	1	0

Table 5: GModel willingness to share its value function for the **cooperative** DoND game mode.

	GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
Tell exact values	0	4	0	9	0	1	0	0	0	0	6	6	5	11
Tell only relative values	15	14	8	11	19	19	1	20	0	0	1	10	7	9
Do not tell preferences	5	0	11	0	1	0	8	0	0	0	12	0	7	0
Do not communicate at all	0	0	1	0	0	0	11	0	20	20	1	4	1	0

Table 6: GModel willingness to share its value function for the **semi-competitive** DoND game mode.

there was no communication between the models we counted it as truthful.

Again, for the cooperative game mode it is not rational to misrepresent one’s value function. Still there are some instances of models doing so. A particularly interesting case for this is the Deepseek-v3.1 model, as it often in the first few messages gave an exact value function but with completely wrong values. Often throughout the course of the game the player would then correct those values and stating the correct value function. Overall all otgher models are generally thoughtful when representing their preferences.

For the semi-competitive game mode it may sometimes be advantageous to lie about one’s preferences to get an advantage in the negotiations. However, other than the Claude 4 model without reasoning, which is misrepresenting its value function slightly more often, the other models remain truthful.

B.5 Qualitative Samples

C Clean Up - Game Details

C.1 Prompt Templates

The prompt templates for both players of Clean Up are given in Figure 14.

Intermittent messages sent by the Game Master to the players are given in Figure 15. Following a player’s response, they are either penalized and reprompted (cf. Figure 15a, 15b), or the turn is passed to the other player (Figure 15c).

C.2 Evaluation Metrics

The Quality Score is calculated by multiplying *Distance Score* with *Penalty Score*.

$$\text{Quality} = \text{Distance Score} \cdot \text{Penalty Score} \cdot 100$$

Distance Score ($DS \in [0, 1]$) is calculated from three components, each representing the sum of Euclidean distances for all identical objects on both grids: the Initial Distance Sum **I** at the start of

TEMPLATE C.1.1

I am your game master, and you are playing a collaborative game with the following grid as game board:

1234567

C	o	o	o				1
o	I	o					2
o	o						3
o	o						4
							5
o					W	T	6
o					o	H	7

* The upper edge displays x-coordinates increasing to the right, and the right edge y-coordinates increasing downward.

* The following objects are randomly placed on your grid: 'W', 'I', 'T', 'C', 'H'.

The other player sees a variation of the game board, where the objects are placed at different random locations. You cannot see the other player's board, and they cannot see yours.

Goal:

Both players need to move the objects on their respective background so that identical objects end up at the same coordinates. You have to communicate with the other player to agree upon a common goal configuration.

Rules:

* In each turn, you can send exactly one of the following two commands:

1. ``SAY: <MESSAGE>``: to send a message (everything up until the next line break) to the other player. I will forward it to your partner.

2. ``MOVE: <OBJECT>, (<X>, <Y>)``: to move an object to a new position, where ``<X>`` is the column and ``<Y>`` is the row. I will inform you if your move was successful or not.

* If you don't stick to the format, or send several commands at once, I have to penalize you.

* If both players accumulate more than 12 penalties, you both lose the game.

* It is vital that you communicate with the other player regarding your goal state! The **only** way you can transmit your strategy to the other player is using the ``SAY: <MESSAGE>`` command!

Moving Objects

* You can only move objects to cells within the bounds of the grid. The target cell must be empty, i.e., it must only contain the symbol 'o'.

* If you try to move an object to a spot that is not empty, or try to move it outside of the grid, I have to penalize you. You get another try.

* Before making a move, double check that the target spot is empty, and does not hold any letter, frame, or line!

End of Game

If you think you reached the goal of aligning all objects, you can ask the other player to finish the game by sending ``SAY: finished?``. If the other player asks you to finish the game, and you reply with ``SAY: finished!``, the game will end.

Both players win if the game ends within 20 rounds, where one round is defined as two players each sending a valid command.

Scoring:

The closer the identical objects are in both game boards, the more points you get. Penalties reduce your points. Can you beat the record?

TEMPLATE C.1.2

Please send a message to the other player to start the game!

TEMPLATE C.1.3

The other player started the game by sending this message:

`"<START_MESSAGE>"`

What is your first command?

(a) Prompt template for Player A in the game Clean Up.

(b) Prompt template for Player B in the game Clean Up.

Figure 14: Clean Up prompt templates for both players

TEMPLATE C.1.4

Penalty: <REASON>

Make sure that your response only contains either SAY: <MESSAGE> or MOVE: <OBJECT>, (<X>, <Y>), and nothing else!

You have collectively accumulated <N> of <M> penalties. Please try again!

<REASON> can be one of the following:

- Your message must not contain anything before the command!
- Your message must not contain anything after the command!
- Your message must not contain anything before or after the command!
- Your message contains more than one command!
- Your message is not in the expected format!
- You must begin the game by sending a message to the other player!

(a) Penalty prompt template for format errors in the game Clean Up

TEMPLATE C.1.5

<REASON>

You have collectively accumulated <N> of <M> penalties. Please try again!

<REASON> can be one of the following:

- Invalid move: (<X>,<Y>) is out of bounds.
- Penalty: (<X>,<Y>) is not empty, but contains '<OBJECT>'.
- Invalid move: Your image has no object with ID '<OBJECT>'.

(b) Prompt templates for invalid moves in the game Clean Up

TEMPLATE C.1.6

<LAST MOVE>

You are currently playing round <R> of maximum <MR>. You have collectively accumulated <N> of <M> penalties. <OTHER PLAYER ACTION>
What is your next command?

<LAST MOVE> can be:

- Your message has been relayed to the other player.
- Moved '<OBJECT>' to (<X>,<Y>) successfully. Your updated grid looks like this:
<GRID>

<OTHER PLAYER ACTION> can be:

- The other player sent this message:
"<MESSAGE>"
- The other player moved an object on their grid.

(c) New turn prompts in the game Clean Up

Figure 15: Intermittent prompts in the game Clean Up

	GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
Misrepresent preferences	0	0	0	1	1	1	0	2	0	0	0	0	0	9
Truthful about preferences	19	20	19	18	19	19	20	18	20	20	20	20	20	11

Table 7: Model truthfulness about its value function for the **cooperative** DoND game mode.

	GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
Misrepresent preferences	0	0	0	2	0	5	0	3	0	0	0	1	1	6
Truthful about preferences	20	18	20	18	20	15	20	17	20	20	20	0	19	14

Table 8: Model truthfulness about its value function for the **semi-competitive** DoND game mode.

the game, the Final Distance Sum $\mathbf{F}$ at the end of the game, and the Expected Distance Sum $\mathbb{E}$ that approximates the distances for randomly placed objects (see Appendix C.4 for details). The Expected Distance Score ES and Distance Reduction Score RS , quantifying how close the players came to the goal of perfect alignment of all objects, are calculated as follows:

$$ES = \max \left\{ 0, 1 - \frac{\mathbf{F}}{\mathbb{E}} \right\}; RS = \max \left\{ 0, 1 - \frac{\mathbf{F}}{\mathbf{I}} \right\}$$

DS is then either 0 if final placement is worse than random, or calculated as the mean of both scores:

$$DS = \begin{cases} \frac{ES+RS}{2} & \text{if } ES > 0 \\ 0 & \text{otherwise} \end{cases}$$

Penalty Score ($PS \in [0.5, 1]$) is calculated from the penalty count P normalized against max. penalties P_m as follows:

$$PS = \frac{P_m}{P - 2P_m} + 1.5$$

We chose the hyperbolic function because it is lenient for low P and harsher for P close to P_m .

C.3 Experimental Setup

Example grids for the three difficulty levels are given in Figure 16.

C.4 Scoring

The expected distance of two objects on an $w \times h$ grid is calculated as follows:

- Assume two independent variables i and j on one discrete dimension $\{1, 2, \dots, w\}$

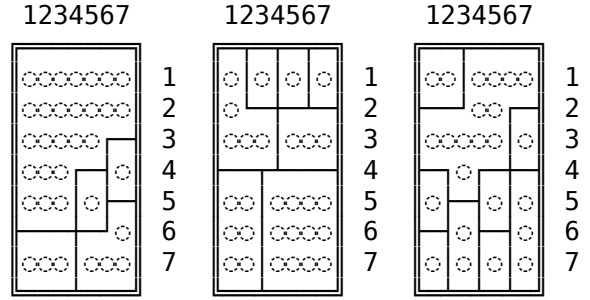


Figure 16: Example grids for *easy*, *medium*, and *hard* levels in Clean Up.

- The expected absolute distance between i and j can be calculated as

$$\mathbb{E}[|i - j|] = \frac{1}{w^2} \sum_{i=1}^w \sum_{j=1}^w |i - j|$$

- The double sum evaluates to

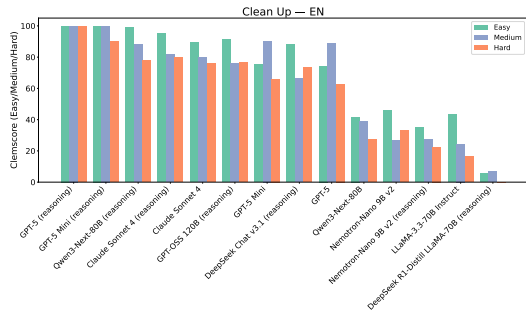
$$\mathbb{E}[|i - j|] = \frac{1}{w^2} \cdot \frac{(w-1)(w+1)}{3} = \frac{w^2 - 1}{3w}$$

For two dimensions, we can then calculate the Euclidean distance of objects o_1 and o_2 as follows:

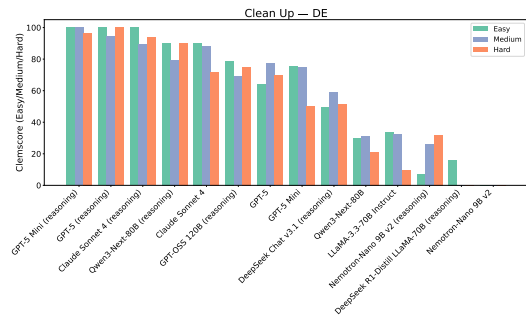
$$\mathbb{E}[|o_1, o_2|] = \sqrt{\left(\frac{w^2 - 1}{3 \times w}\right)^2 + \left(\frac{h^2 - 1}{3 \times h}\right)^2}$$

For a 7×7 grid, we thus get an expected distance of ≈ 4.19 for two randomly placed objects. To calculate the Expected Distance Sum, we simply multiply it by the number of objects.

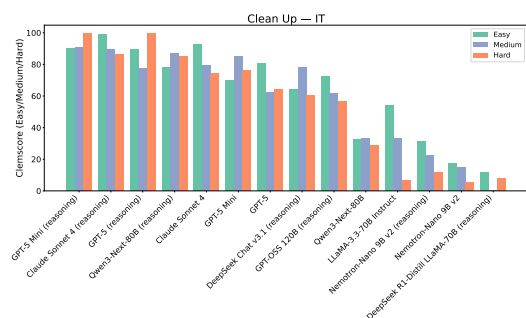
This is only an approximation, since we neither take into account the actual grid configuration, nor that two objects cannot have the same location on the same grid.



(a) Results for each experiment for English.



(b) Results for each experiment for German.



(c) Results for each experiment for Italian.

Figure 17: Comparison of experimental results for English (top), German (middle), and Italian (bottom).

C.5 Overall Results

C.5.1 Overall Results across levels

As explained in section ??, we designed three broad game difficulty levels based on the ratio of empty cells in a grid. We hypothesize that the denser the background grid is, the more likely the target locations clash with occupied cells, and therefore the more likely models get penalized in scoring. In addition, at each difficulty level, there are three sub-levels: either 3, 5, or 7 objects are placed on the grid, on the assumption that the more objects there are, the longer the gameplay traces are, and the less likely that models achieve a high score with mere luck.

Figure 18 plots the main score; clear stratifica-

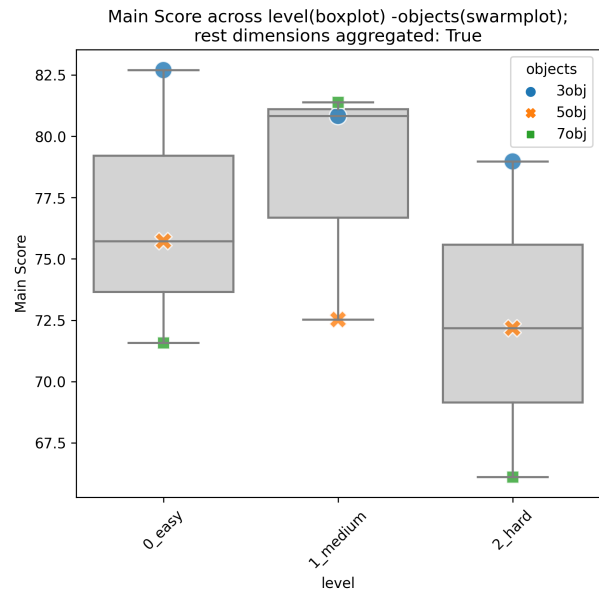


Figure 18: Main Score across levels and sub-levels

tions of performance across the number of objects are observed in easy and hard levels. When we break down the Main Score into Distance Score and Penalty Score in Figure 19, the stratification at the medium level can also be found.

Games with 3 objects achieve the highest scores, regardless of background empty cell ratios, which corroborates our assumption that games with a low number of objects are easier. However, levels with 5 objects have the lowest Distance Score in the medium level and the lowest Penalty Score in both easy and hard levels, indicating that there are factors other than the number of objects that affect performance.

C.5.2 Overall Results across model properties

Figure 20 illustrates that the Main Scores of commercial models are skewed toward the upper end of the scale, with a clear concentration in the 75100 range, particularly under reasoning mode. In contrast, open models generally score lower, with many results in the 050 range. Notably, the dots are much sparser for open models when reasoning is on. This is because the Main Score is not recorded when the maximum number of penalties is reached; in these cases, the game will be marked as aborted.

In Figure 21, we further investigate the distribution of Distance Score and Penalty Score. The aborted games, while not shown in the Main Score plot, are represented here with an "x" marker. We set the Penalty Score to 0.5 when the models reach the maximum allowed penalties. If they incur one

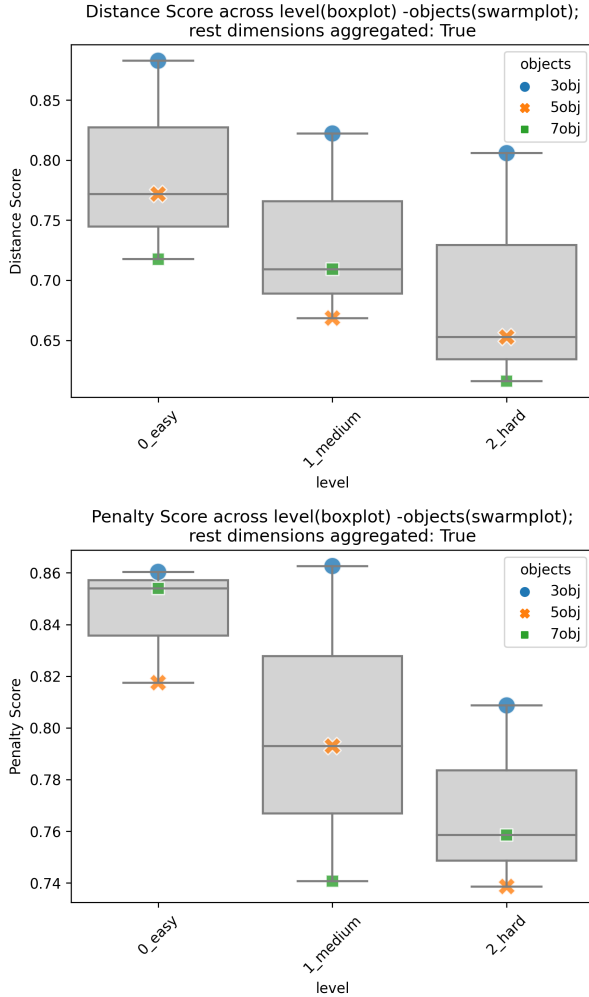


Figure 19: Sub Scores across levels and sub-levels

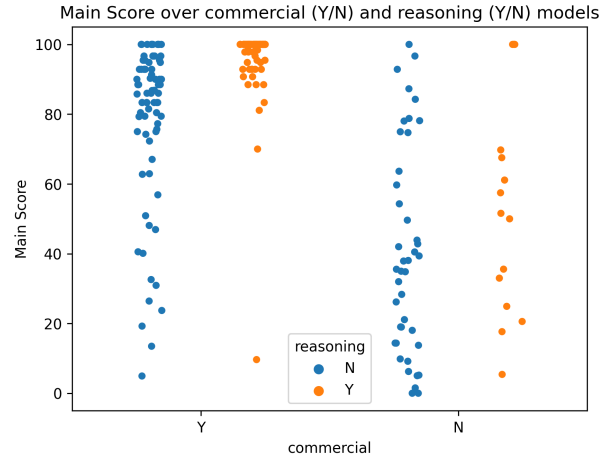


Figure 20: Main Score over different model properties

more penalty, the game is marked as aborted, and the Main Score won't be recorded.

In the Penalty Score subplot, all game plays with Penalty Score < 0.4 are marked with "x" by definition. While the total number of game plays in each of the four lanes is the same, we see that the open reasoning models have the highest share of aborted games, followed by open non-reasoning models. In the Distance Score subplot, we see that the aborted games span widely for open non-reasoning models, indicating some game plays, although achieving a high Distance Score, have pitifully accumulated too many penalties and thus have been marked aborted and taken out of the account of the Main Score. In contrast, commercial models have generally better Penalty Score and Distance Score.

An interesting observation is that, in the Distance Score subplot, the aborted games tend to reside in lower ranges for non-reasoning models, while the reasoning models might play an aborted game despite achieving a high Distance Score. This is especially prominent for open non-reasoning models, and hints that the reasoning mode, rather than helping in games where these models might achieve a high Distance Score, might introduce noises that lead to more penalties.

C.6 Detailed Analysis

Idea: We hypothesize that Instruction Following ability and Spatial Estimation abilities serve as the foundational pillars of performance, as illustrated in Figure 22. With weaker pillars, model reasoning might not add much to the performance.

Penalties include the Invalid Format Penalty and the Invalid Move Penalty. We plan to use the ratio of Invalid Format Penalty as an agent of Instruction

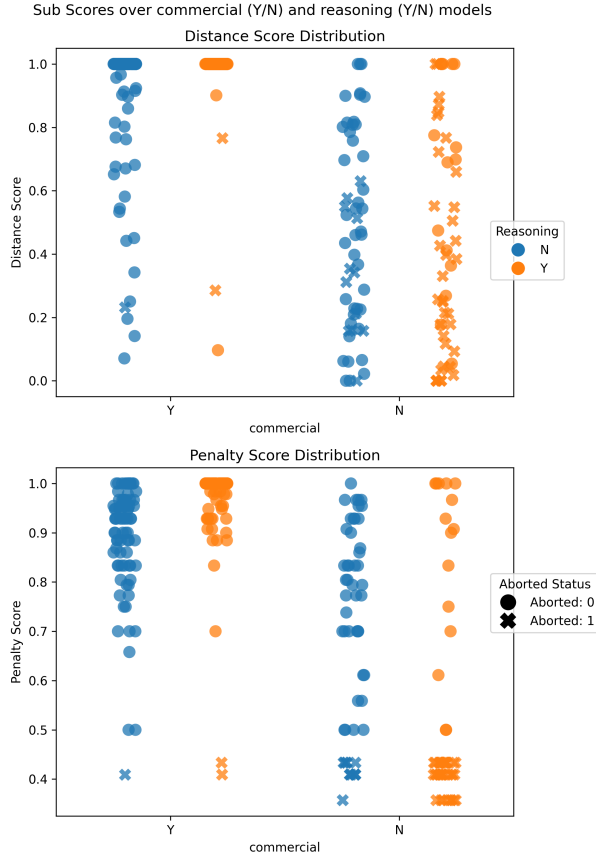


Figure 21: Sub Scores over different model properties

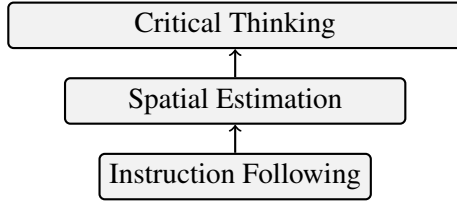


Figure 22: Capability Hierarchical

Following ability, sort the models by it, compare the result of the same model with reasoning on or off, and see what percentage the reasoning capability adds to the performance. The assumption is that models with better Instruction Following abilities benefit more from the introduction of the reasoning mode.

We plan to apply the same analysis to Invalid Move Penalty.

C.7 Qualitative Samples

Idea: We plan to do qualitative analysis from the following four angles: Spatial Estimations, Strategy, Critical Thinking, Communication & Collaboration, and analyze and compare performance across commercial models (Y/N) and reasoning models (Y/N). For GPT-5, we compare across

model size (original / mini) as well.

Here we briefly motivate each of these angles. Concrete examples and analysis will be added later.

C.7.1 Spatial Estimation

From this angle, we check if models can propose valid moving plans. It captures models' Spatial Estimation ability in our grid settings and serves as a foundational pillar of the overall performance. When models are not capable in this regard, they tend to propose invalid moving plans, and attempts to resolve the wrong movements after getting penalties often conduce to cascades of invalid moving plans, therefore quickly exhaust the number of allowed penalties and end up either aborting the game, or attain a low penalty score.

C.7.2 Strategy

In addition to the crude Spatial Estimation, we check how models come up with moving plans. We have observed some smart strategies and their not-so-smart counterparts. In some game plays, one model uses the initial coordinates of the objects to instruct the other model as target locations, thus avoiding penalties of moving objects to cells occupied by the background grids to a large extent; while in some other game plays, models appear to exchange initial coordinates just for the sake of exchanging, without making use of this information. Another example is in some game plays, models choose the most sparse row as the target, and lay objects one by one on it; while in some other game plays, the models unnecessarily chose geometric patterns to lay the objects, such as on the diagonal, disregarding the occupied cells, and run into complexities that they could have dodged.

C.7.3 Critical Thinking

Some models show the ability to critically reason about the moving plan that either itself or the other player proposed. They either refute the plan immediately when it's proposed, or take a sharp brake when they are just about to execute the invalid moving plan. This is, as expected, more frequent in reasoning models compared to non-reasoning models. However, on some occasions, we see pseudo-reasoning: a model rejects a correct plan, utters reasoning-like sentences, and proposes another plan that sometimes is correct, sometimes is wrong.

C.7.4 Communication & Collaboration

The above three dimensions positively correlate with the model performance: The better the models are at them, the higher the scores. This dimension, however, is different. Specifically, we observe how models communicate with each other to recover from invalid moves due to target locations being occupied. Some models go at lengths to execute their alternative plans without letting their partner know, they might still have a good score, simply because both player 1 and player 2 adopt the exact same strategy, for example, move the object to one spot next to the original target location. This leads to inflated scores that can be debunked when two models of different default recovery strategies, or one model and one human, play against each other.

D Air Balloon Survival - Game Details

D.1 Prompt Templates

The prompt template for both players of the Hot Air Balloon Survival game is given in Figure 23. Parse error prompts handed to players when their responses did not follow the format instructions are given in Figure 24, and game error prompts for invalid actions are given in Figure 25.

Substrings Figure 23 marked by \$. . . \$ are placeholders which get replaced depending on the game instance. Note that the last three placeholders may also be replaced by the empty string depending on the game settings. Here we list the replacements for placeholders in the initial prompt:

- \$LIMIT\$ maximum weight for a deal.
- \$ITEM_WEIGHTS\$ mapping from items to weights.
- \$UTILITY_SCALE_PLAYER\$ mapping from items to preference values for a given player.
- \$STRATEGIC_REASONING_FORMAT\$
STRATEGIC REASONING: {'. . .'}
Describe your strategic reasoning or anticipation explaining your choice of action. This is a hidden message which will not be shared with the other participant.
- \$REQUIRE_ARGUMENT\$ You must include the ARGUMENT format at least once somewhere in all of your messages.
- \$STRATEGIC_REASONING_RULE\$

You must include the STRATEGIC REASONING format only once at the very beginning of every one of your messages and not more often. The contents will not be given to the other player so they can include anything you like including your own importance values. Here you should reason step by step to come up with your next move.

D.2 Evaluation Metrics

Each player $p \in P$ receives an individual score based on the utility of the final deal D , i.e. the sum of the preference values of items in D . This sum gets normalized by the value of the optimal solution to their instance of the 0/1 Knapsack Problem. The overall game score is defined as the harmonic mean of the two players normalized scores, further normalized by the maximum harmonic mean achievable for an instance. If the total weight of the deal exceeds the capacity W , the score is set to 0. We chose the harmonic mean as our main metric as it rewards collectively balanced outcomes by favouring deals where both players achieve high scores and penalizing those where one player benefits disproportionately. The individual scores of players is each players outcome relative to the optimal solution of their own knapsack problem. For a final deal $D \subseteq \{1, \dots, n\}$, the normalized score of player $p \in P$ is

$$f_p(D) = 100 \cdot \frac{\sum_{i \in D} v_p(i)}{\text{OPT}_p},$$

where OPT_p is the value of the optimal solution to player p 's 0/1 Knapsack Problem. The harmonic mean of the players' normalized scores is defined as

$$f_{\text{harm}}(D) = \begin{cases} \frac{2f_1(D)f_2(D)}{f_1(D)+f_2(D)} & \text{if } \sum_{i \in D} w_i \leq W, \\ 0 & \text{otherwise.} \end{cases}$$

The final game score is the normalized harmonic mean,

$$\text{Quality}(D) = 100 \cdot \frac{f_{\text{harm}}(D)}{\text{OPT}^*},$$

where OPT^* denotes the maximum harmonic mean achievable in the given instance.

TEMPLATE D.1.1

You are participating in a collaborative negotiation game.

Together with another participant, you must agree on a single set of items that will be kept. Each of you has your own view of how much each item matters to you (importance). You do not know how the other participant values the items. Additionally, you are given the effort each item demands. You may only agree on a set if the total effort of the selected items does not exceed a shared limit:

LIMIT = \$LIMIT\$

Here are the individual item effort values:

Item effort = \$ITEM_WEIGHTS\$

Here is your personal view on the importance of each item:

Item importance values = \$UTILITY_SCALE_PLAYER\$

Goal:

Your goal is to negotiate a shared set of items that benefits you as much as possible (i.e., maximizes total importance to YOU), while staying within the LIMIT. You are not required to make a PROPOSAL in every message - you can simply negotiate as well. All tactics are allowed!

Interaction Protocol:

You may only use the following structured formats in a message:

PROPOSAL: {'A', 'B', 'C', }
Propose keeping exactly those items.

REFUSE: {'A', 'B', 'C', }
Explicitly reject opponent's proposal.

ARGUMENT: {'...'}
Defend your last proposal or argue against the player's proposal.

AGREE: {'A', 'B', 'C', }
Accept the opponent's proposal which ends the game.

\$STRATEGIC_REASONING_FORMAT\$

Rules:

You may only AGREE on a proposal the other party has logged via PROPOSAL.

You may only REFUSE a proposal the other party has logged via PROPOSAL.

Total effort of any PROPOSAL or AGREE set must be $\leq$ LIMIT.

Do NOT reveal your hidden importance scores.

A tag in a structured format must be followed by colon and whitespace. The argument must be a python set containing 0 or more strings.

So, it must be of the form TAG: {...}

Strictly follow the interaction protocol and DO NOT write anything beyond the given structure.

The game ends when one side gives an AGREE to a PROPOSAL made by the other player.

The content in your response which can be handed to the other player has to be non-empty.

Only proposals which have been logged via the PROPOSAL format structure and which haven't been refused via REFUSE are active.

\$REQUIRE_ARGUMENT\$

\$STRATEGIC_REASONING_RULE\$

Figure 23: Initial prompt template of Hot Air Balloon. Note that for the second player we append newline and the phrase 'You will now receive the first message of the other player.' at the very end of the initial prompt.

TEMPLATE D.1.2

Your response did not start with the proper strategic reasoning tag at the very beginning of your response.
The very first structured format must be of the form STRATEGIC REASONING: {...}. Try again.

(a) Parse error prompt in case strategic reasoning tag at the beginning of a response is missing. Only applies when strategic reasoning tag is required

TEMPLATE D.1.3

Your response did not contain an argument.
You must include the structured format ARGUMENT: {...} somewhere in your response. Try again.

(b) Parse error prompt in case an argument is missing. Only applies when argument tag is required

TEMPLATE D.1.4

Your response contained an untagged sequence or you used STRATEGIC REASONING more than once.
You may only use the structured formats as explained in the initial message.
They must all be of the form TAG: {...}.

(c) Parse error prompt in case the response contains an untagged sequence.

TEMPLATE D.1.5

Your response only contained a strategic reasoning tag.
You must at least include one more valid tag in your response, so that the other player receives a message. Try again.

(d) Parse error prompt in case response only contained a strategic reasoning tag. Only applies when argument tag is not required.

TEMPLATE D.1.6

You used a PROPOSAL tag, but did not provide a valid python set containing strings as arguments, e.g. {'A', 'B', 'C', ...}. Try again.

(e) Parse error prompt in case of an invalid python set as argument.

Figure 24: Parse error prompts handed to player when a response did not follow the instructions on structured formats.

TEMPLATE D.1.7

You refused a proposal which is not active.
Proposals are only active if they have been
logged by the other player via PROPOSAL and
have not been deactivated by you via REFUSE.
Try again.

(a) Game error prompt in case a non-active was refused.

TEMPLATE D.1.8

You made more than one agreement.
Final deals cannot be ambiguous. Try again.

(b) Game error prompt in case agreement to a deal was ambiguous.

TEMPLATE D.1.9

You agreed to a proposal which is not active.
Proposals are only active if they have been
logged by the other player via PROPOSAL and
have not been deactivated by you via REFUSE.
Try again.

(c) Game error prompt in case .

TEMPLATE D.1.10

Your proposal includes items which are not in
the game. Try again.

(d) Game error prompt in case a non-active proposal was agreed to.

Figure 25: Game error prompts handed to player when a response did not follow the instructions on the rules of the game.

Model	Pareto Adherence
GPT-OSS 120B (reasoning)	0.763
GPT-5 (reasoning)	0.675
GPT-5 Mini (reasoning)	0.548
Qwen3-Next-80B (reasoning)	0.464
Nemotron-Nano 9B v2 (reasoning)	0.367
GPT-5 Mini	0.269
GPT-5	0.217
Claude Sonnet 4 (reasoning)	0.164
DeepSeek Chat v3.1	0.150
Claude Sonnet 4	0.011
LLaMA-3.3-70B Instruct	0.000
Nemotron-Nano 9B v2	0.000
DeepSeek R1-Distill LLaMA-70B	-
Qwen3-Next-80B	-

Table 9: Average per instance Pareto adherence rate for proposals per model. Values were computed on English instances with 15 items.

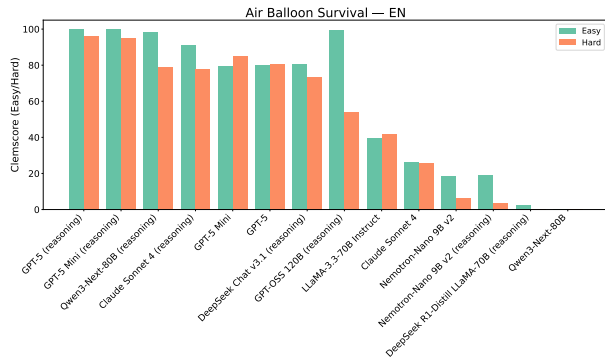
Model	Alternation
DeepSeek R1-Distill LLaMA-70B (reasoning)	1.000
GPT-5 Mini (reasoning)	1.000
GPT-5	1.000
Claude Sonnet 4	1.000
GPT-5 (reasoning)	1.000
GPT-5 Mini	1.000
GPT-OSS 120B	0.991
Claude Sonnet 4 (reasoning)	0.947
Qwen3-Next-80B (reasoning)	0.823
Nemotron-Nano 9B v2	0.875
DeepSeek Chat v3.1	0.734
LLaMA-3.3-70B Instruct	0.573
Nemotron-Nano 9B v2 (reasoning)	0.167
Qwen3-Next-80B	-

Table 10: Average alternation rate per model on all English instances.

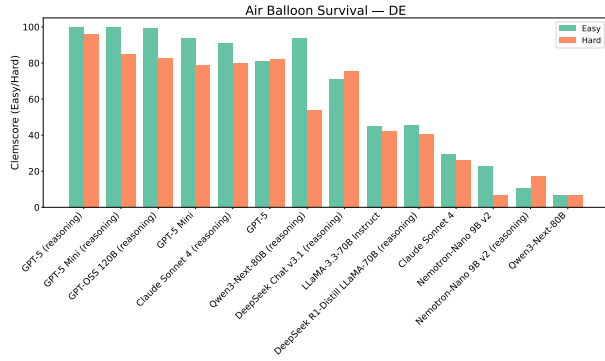
D.3 Overall Results**D.4 Detailed Analysis****D.4.1 Negotiation Dynamics****D.4.2 Collaboration****D.4.3 Critical Thinking****D.5 Qualitative Samples**

Model	P1	P2
GPT-OSS 120B	96.314	74.077
Nemotron-Nano 9B v2 (reasoning)	92.475	64.105
GPT-5 (reasoning)	89.634	77.853
GPT-5 Mini (no reasoning)	86.698	57.792
Qwen3-Next-80B (reasoning)	85.928	75.960
GPT-5 Mini (reasoning)	85.891	79.753
GPT-5 (no reasoning)	85.010	63.461
DeepSeek Chat v3.1	79.987	56.153
Claude Sonnet 4 (no reasoning)	77.342	64.821
Claude Sonnet 4 (reasoning)	70.231	72.229
Nemotron-Nano 9B v2 (no reasoning)	56.113	61.660
LLaMA-3.3-70B Instruct	52.536	40.910
DeepSeek R1-Distill LLaMA-70B	-	-
Qwen3-Next-80B	-	-

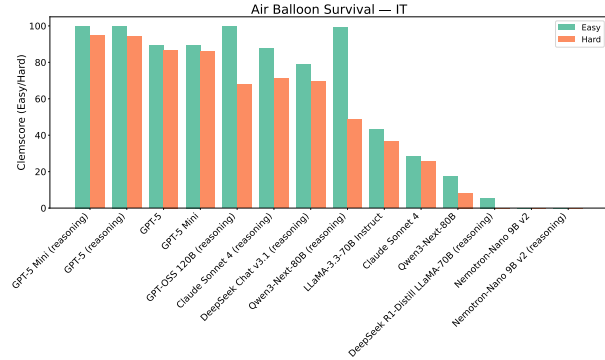
Table 11: Average player scores on English opposing goals instances (only considering played games).



(a) Clem scores on English instances.



(b) Clem scores on German instances.

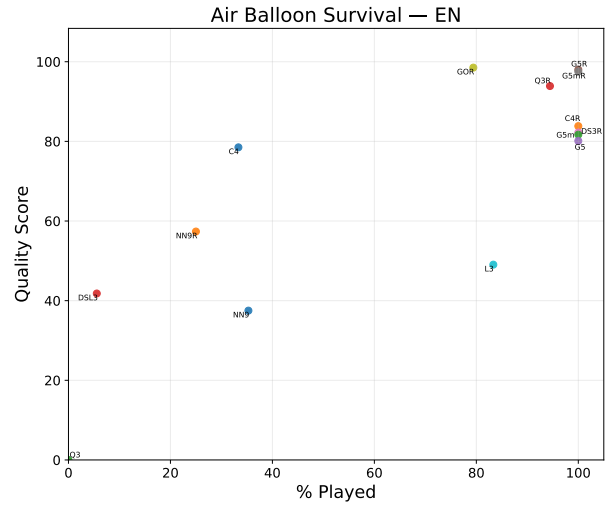


(c) Clem scores on Italian instances.

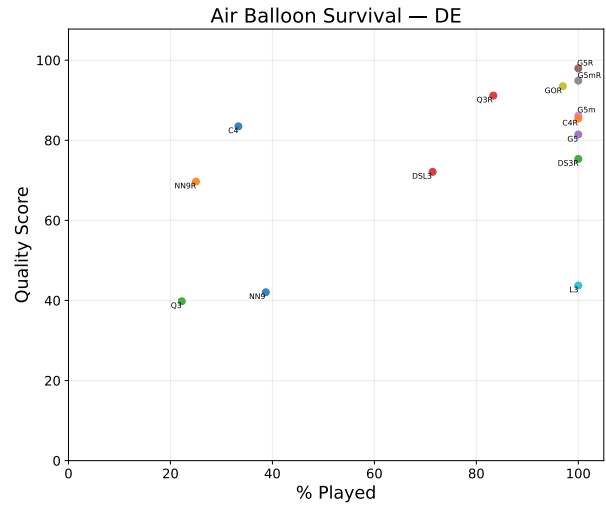
Figure 26: Overall model performance for easy and hard sections of experiments for all languages.

Model	Proactive	Reactive	Total G
Claude 4 (no reasoning)	0	13	13
Claude 4 (reasoning)	1	13	14
DeepSeek Chat v3.1	11	12	23
DS-R1-Distill LLaMA-70B	1	0	1
GPT-5 (no reasoning)	15	5	21
GPT-5 (reasoning)	17	3	24
GPT-5 Mini (no reasoning)	19	1	21
GPT-5 Mini (reasoning)	19	5	26
GPT-OSS 120B	5	6	21
LLaMA-3.3-70B Instruct	3	10	19
Nemotron 9B (no reasoning)	21	1	22
Nemotron 9B (reasoning)	0	2	2
Qwen3-Next-80B	4	5	15

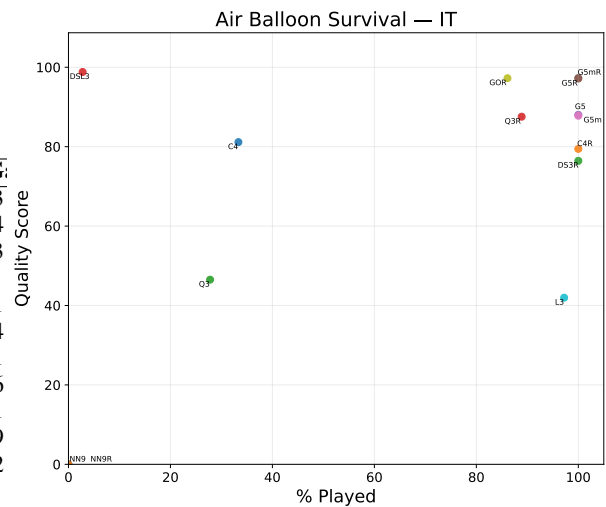
Table 12: Initial reasoning style - classified by counting collaborative (proactive) versus egocentric (reactive) keywords in the first turn of each game within Strategic Reasoning tag. Calculated on all English instances.



(a) English

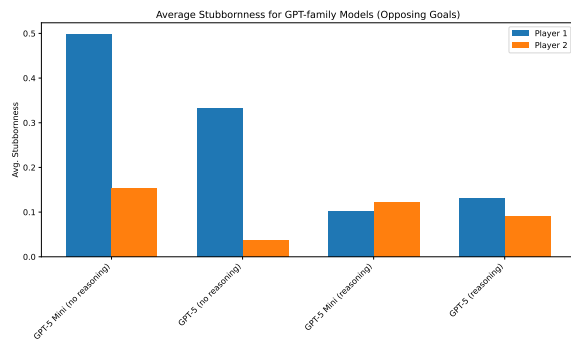


(b) German

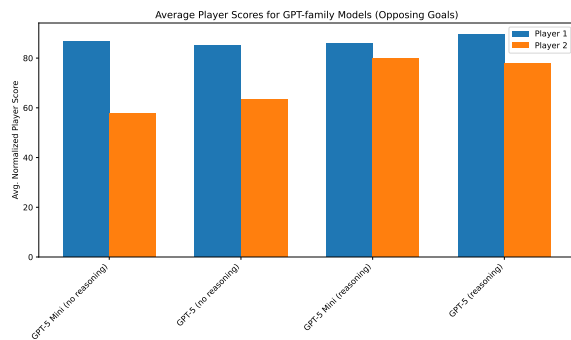


(c) Italian

Figure 27: Quality and Played scores on *Air Balloon Survival* instances for each language. Scatters are annotated with acronyms of model names. ‘R’ denotes reasoning model.

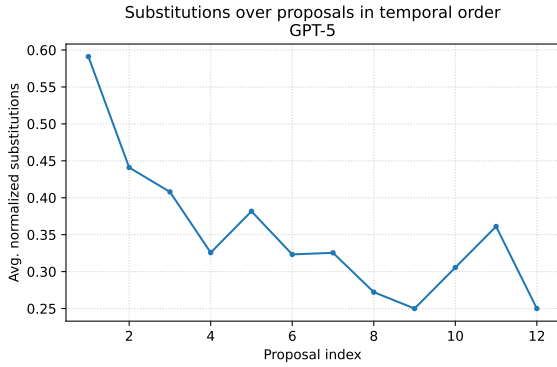


(a) Average stubbornness per player per instance for reasoning vs. conventional GPT-family models.

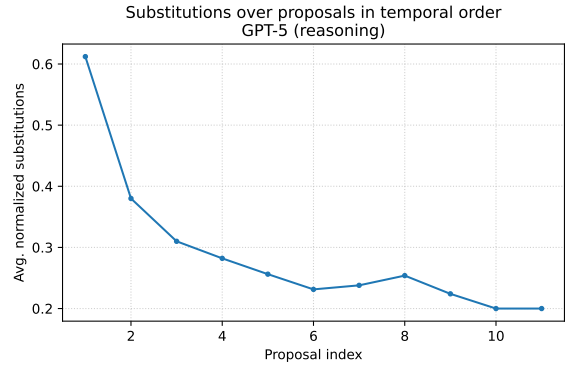


(b) Average player scores for GPT-family models.

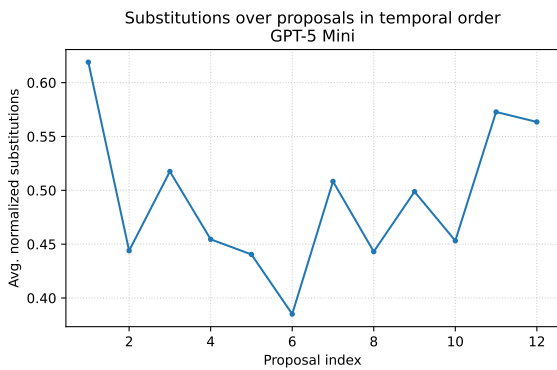
Figure 28: Comparison of player 1 and player 2 in GPT-family models. Values were computed on English instances with opposing-goals setting



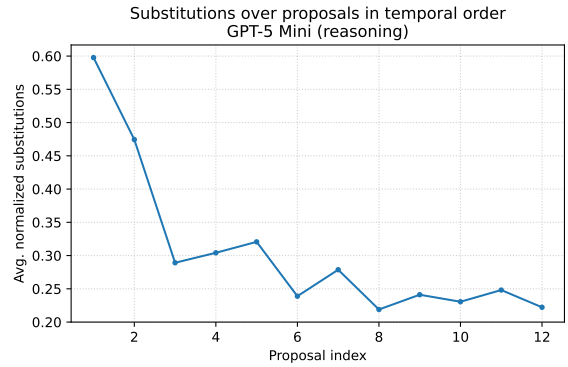
(a) GPT-5 (no reasoning). Curve is decreasing, but less smoothly compared to the reasoning version.



(b) GPT-5 (reasoning). Very smooth and decreasing curve, indicating coherent bargaining and stronger early disagreement.



(c) GPT-5 Mini (no reasoning). Substitutions rise in longer negotiations, reflecting frustration and less coherent bargaining.



(d) GPT-5 Mini (reasoning). Substitutions decrease on average, showing more coherent bargaining than non-reasoning model.

Figure 29: Comparison of bargaining coherence across models. Reasoning in top models amplifies the smoothed negotiation, with each proposal building on the previous. Results are from English experiments, capped at 12 proposals.